

Round 3 - Signal Tracking: Real-Time Analysis

Competition: Data Vortex | AARUUSH'26 **Theme:** Rebuilding the Social Engine **Team:** Entropy **Members:** Saanvi Grover & Aditya Sharma

Assigned topic: Public Reaction to a Major Software Update **Event tracked:** the **iOS 27 / iPadOS 27 / macOS 27** release (14 September 2026), with **Android 17** (18 September 2026) as a comparison release.

This notebook loads the self-collected dataset produced by `Entropy_collect_data.py`, applies the **Round 2 sentiment model** unchanged, and then answers the four questions the round asks for:

1. **Activity** - when did people post, and where are the spikes?
2. **Sentiment** - how did the balance of opinion move, and when did it shift?
3. **Topics and entities** - what exactly are people talking about?
4. **Triggers** - which real events explain the spikes and shifts?

```
In [1]: import os, sys, json, re, warnings, time
from pathlib import Path
warnings.filterwarnings("ignore")
os.environ["HF_HUB_OFFLINE"] = "1"
os.environ["TOKENIZERS_PARALLELISM"] = "false"

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.dates as mdates
from matplotlib.ticker import PercentFormatter
from scipy import stats
from sklearn.decomposition import NMF
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix, f1_score
from sklearn.feature_extraction.text import TfidfVectorizer, CountVectorizer

ROOT = Path.cwd()
DATA, IMG, OUT = ROOT / "data", ROOT / "images", ROOT / "outputs"
for p in (IMG, OUT):
    p.mkdir(exist_ok=True)
R2 = ROOT.parent / "Round2-Semantic-Recovery"
sys.path.insert(0, str(R2))
import Entropy_nlp_utils as nu # the Round 2 module, unchanged

SEED = 42
N_BOOT = 2000
np.random.seed(SEED)
RESCORE = os.environ.get("ENTROPY_RESCORE", "0") == "1"
SCORED = OUT / "Entropy_round3_scored_posts.csv"
LAUNCH = {"iOS 27": pd.Timestamp("2026-09-14 17:00", tz="UTC"), "Android 17": pd.Timestamp("2026-09-18 17:00", tz="UTC")}
METRICS = {}

SURF, INK, INK2, MUTED, GRID, BASE = "#fcfcfb", "#0b0b0b", "#52514e", "#898781", "#e1e0d9", "#c3c2b7"
SLOTS = ["#2a78d6", "#eb6834", "#1baf7a", "#eda100", "#e87ba4"]
S1, S2 = SLOTS[0], SLOTS[1]
SENT = {"Negative": "#e34948", "Neutral": "#8f8d86", "Positive": "#2a78d6"}
plt.rcParams.update({
    "font.family": ["Segoe UI", "DejaVu Sans"], "font.size": 9.5, "figure.dpi": 110,
    "figure.facecolor": SURF, "axes.facecolor": SURF, "savefig.facecolor": SURF, "savefig.dpi": 200,
    "axes.edgecolor": BASE, "axes.labelcolor": INK2, "xtick.color": MUTED, "ytick.color": INK2,
    "axes.grid": True, "grid.color": GRID, "grid.linewidth": 0.6, "axes.axisbelow": True,
    "legend.frameon": False, "axes.spines.top": False, "axes.spines.right": False,
    "axes.titlesize": 11, "axes.titleweight": "bold", "axes.titlelocation": "left", "axes.titlecolor": INK,
})
def save(fig, name):
    fig.savefig(IMG / f"Entropy_r3_{name}.png", bbox_inches="tight")
```

```
plt.show()
print("Round 2 module:", nu.__file__)
```

Round 2 module: C:\Users\Saanvi\Desktop\Data Vortex\Round2-Semantic-Recovery\Entropy_nlp_utils.py

1. Data collection method and dataset structure

The collector (`Entropy_collect_data.py`) is a plain-Python script using only free, key-less public endpoints. The manifest it writes records exactly what was requested and when.

```
In [2]: manifest = json.loads((DATA / "Entropy_round3_collection_manifest.json").read_text(encoding="utf-8"))
raw = pd.read_csv(DATA / "Entropy_round3_collected_posts.csv")
news = pd.read_csv(DATA / "Entropy_round3_news_timeline.csv")
# parse_dates can silently leave mixed-offset timestamps as strings, so convert explicitly
raw["created_utc"] = pd.to_datetime(raw.created_utc, format="mixed", utc=True)
news["published_utc"] = pd.to_datetime(news.published_utc, format="mixed", utc=True)
keys = ["topic", "started_utc", "finished_utc", "requests", "failed_requests", "rows_raw",
        "rows_fetched_this_run", "rows_final", "append_mode", "reddit_popularity_pass",
        "unique_authors", "news_articles", "time_span"]
print(json.dumps({k: manifest[k] for k in keys if k in manifest}, indent=2))
structure = pd.DataFrame({
    "endpoint": ["www.reddit.com/*/search.rss, /comments/*.rss", "hn.algolia.com/api/v1/search",
                "{instance}/api/v1/timelines/tag/{tag}", "{instance}/api/v3/search",
                "news.google.com/rss/search"],
    "auth": ["none (anonymous Atom)", "none", "none", "none", "none"],
    "engagement field": ["not exposed by RSS", "points, num_comments", "favourites + reblogs, replies",
                        "score, comments", "n/a (event timeline)"],
}, index=["reddit", "hackernews", "mastodon", "lemmy", "googlenews"])
display(structure)
display(raw.groupby(["source", "kind"]).size().unstack(fill_value=0))
display(raw[["source", "source_detail", "kind", "created_utc", "title", "text", "score", "n_replies",
               "query"]].head(4))
```

```
{
  "topic": "Public reaction to a major software update: iOS 27 / iPadOS 27 / macOS 27, with Android 17 as com
parison",
  "started_utc": "2026-09-20T10:33:04.228439+00:00",
  "finished_utc": "2026-09-20T11:42:12.724645+00:00",
  "requests": 199,
  "failed_requests": 19,
  "rows_raw": 9861,
  "rows_fetched_this_run": 6379,
  "rows_final": 5498,
  "append_mode": true,
  "reddit_popularity_pass": 12,
  "unique_authors": 3271,
  "news_articles": 388,
  "time_span": {
    "first": "2026-09-08 01:10:09+00:00",
    "last": "2026-09-20 11:38:50+00:00"
  }
}
```

	endpoint	auth	engagement field
reddit	www.reddit.com/*/search.rss, /comments/*.rss	none (anonymous Atom)	not exposed by RSS
hackernews	hn.algolia.com/api/v1/search	none	points, num_comments
mastodon	{instance}/api/v1/timelines/tag/{tag}	none	favourites + reblogs, replies
lemmy	{instance}/api/v3/search	none	score, comments
googlenews	news.google.com/rss/search	none	n/a (event timeline)

	kind	comment	post
source			
hackernews	1919	126	
lemmy	29	88	
mastodon	0	1594	
reddit	688	1054	

	source	source_detail	kind	created_utc	title	text	score	n_replies	query
0	hackernews	story	post	2026-09-08 01:10:09+00:00	Show HN: Browse 27 years of movie ticket stubs...	This is one of those projects that's been...	2.0	0.0	iOS 27
1	hackernews	story	post	2026-09-08 01:41:59+00:00	I killed every houseplant I owned, so I built ...	I've killed more houseplants than I'...	2.0	3.0	Android 17
2	hackernews	comment	comment	2026-09-08 02:19:03+00:00		URL? Some photos of the corpses? Indoor or bal...	NaN	NaN	comment_thread
3	reddit	r/apple	comment	2026-09-08 02:38:23+00:00		I really am itching for a new form factor iPho...	NaN	NaN	comment_thread

```
In [3]: def ascii_ratio(s):
s = str(s)
return sum(ch.isascii() for ch in s) / max(len(s), 1)

BOILERPLATE = re.compile(r"submitted by\s*/?u/?\S*|\[link\]|\[comments\]|/u/\w+|/r/\w+", re.IGNORECASE)
def strip_boilerplate(t):
t = BOILERPLATE.sub(" ", str(t))
t = re.sub(r"https?://\S+", " ", t)
return re.sub(r"\s+", " ", t).strip()

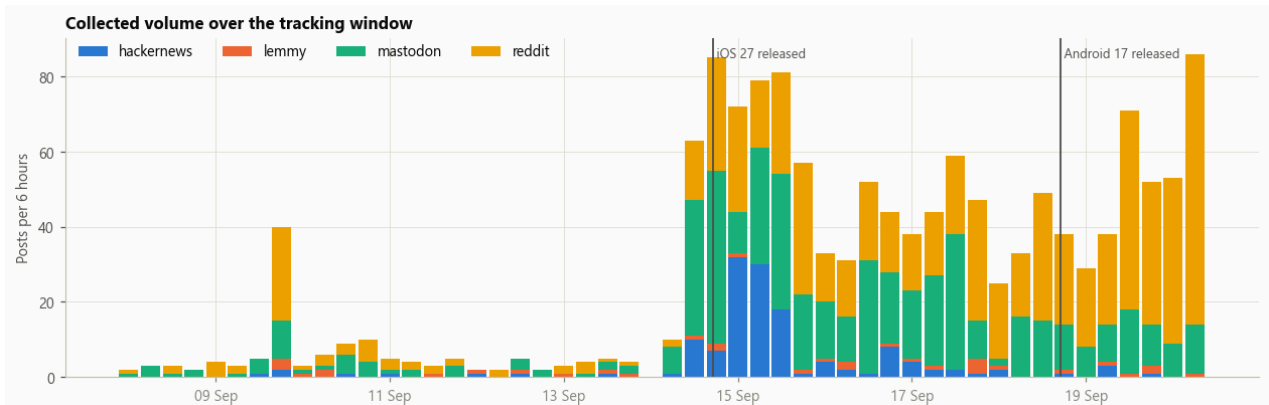
df = raw.copy()
# Reddit's Atom feed appends "submitted by /u/x [link] [comments]" to every entry; that
# footer is not opinion text and would otherwise dominate the word-level analysis.
df["text_full"] = df.text_full.astype(str).map(strip_boilerplate)
df = df[df.text_full.str.len() >= 15]
df["is_english"] = (df.lang.fillna("").eq("en") | df.lang.isna() | df.lang.fillna("").eq("")) &
(df.text_full.map(ascii_ratio) > 0.9)
ENTITIES = {
    "iOS 27": r"\bios\s*27\b|\bios27\b",
    "iPadOS 27": r"\bipados\s*27\b",
    "macOS 27": r"\bmacos\s*27\b|\bmac\s*os\s*27\b",
    "Apple Intelligence": r"apple intelligence|\bsiri\b",
    "Android 17": r"\bandroid\s*17\b|\bandroid17\b",
}
for name, pat in ENTITIES.items():
    df[name] = df.text_full.str.lower().str.contains(pat, regex=True, na=False)
df["on_topic"] = df[list(ENTITIES)].any(axis=1)
df["apple_side"] = df[["iOS 27", "iPadOS 27", "macOS 27", "Apple Intelligence"]].any(axis=1)
df["android_side"] = df["Android 17"]
work = df[df.on_topic & df.is_english].copy().reset_index(drop=True)
METRICS["dataset"] = {
    "collected_rows": int(len(df)), "on_topic_rows": int(df.on_topic.sum()),
    "on_topic_english_rows": int(len(work)), "unique_authors": int(work.author_hash.nunique()),
    "window_start": str(work.created_utc.min()), "window_end": str(work.created_utc.max()),
    "span_hours": round(float((work.created_utc.max() - work.created_utc.min()).total_seconds() / 3600),
1),
    "by_source": work.source.value_counts().to_dict(), "by_kind": work.kind.value_counts().to_dict(),
    "entity_counts": {k: int(work[k].sum()) for k in ENTITIES},
}
print(json.dumps(METRICS["dataset"], indent=2))
```

```
{
  "collected_rows": 5491,
  "on_topic_rows": 1781,
  "on_topic_english_rows": 1403,
  "unique_authors": 1132,
  "window_start": "2026-09-08 03:48:27+00:00",
  "window_end": "2026-09-20 10:44:03.162000+00:00",
  "span_hours": 294.9,
  "by_source": {
    "reddit": 718,
    "mastodon": 513,
    "hackernews": 138,
    "lemmy": 34
  },
  "by_kind": {
    "post": 1250,
    "comment": 153
  },
  "entity_counts": {
    "iOS 27": 784,
    "iPadOS 27": 83,
    "macOS 27": 378,
    "Apple Intelligence": 344,
    "Android 17": 127
  }
}
```

2. Time window

The collection window starts a week before the iOS 27 release so that a pre-launch baseline exists, and runs to the moment of collection. Posting time is taken from each platform's own timestamp, and everything is normalised to UTC.

```
In [4]: hourly = work.set_index("created_utc").resample("1h").size().rename("posts").to_frame()
daily = work.set_index("created_utc").resample("1D").size()
fig, ax = plt.subplots(figsize=(11.5, 3.8))
pivot = work.pivot_table(index=pd.Grouper(key="created_utc", freq="6h"), columns="source",
values="post_id", aggfunc="count").fillna(0)
bottom = np.zeros(len(pivot))
for src, c in zip(pivot.columns, SLOTS):
    ax.bar(pivot.index, pivot[src], width=0.22, bottom=bottom, color=c, label=src)
    bottom += pivot[src].values
for label, ts in LAUNCH.items():
    if pivot.index.min() <= ts <= pivot.index.max():
        ax.axvline(ts, color=INK2, lw=1.2)
        ax.text(ts, ax.get_ylim()[1] * 0.97, f" {label} released", fontsize=8.5, color=INK2, va="top")
ax.xaxis.set_major_formatter(mdates.DateFormatter("%d %b"))
ax.set_ylabel("Posts per 6 hours"); ax.set_title("Collected volume over the tracking window")
ax.legend(ncol=4, loc="upper left", bbox_to_anchor=(0, 1.02))
fig.tight_layout()
save(fig, "volume_timeline")
display(daily.rename("posts per day").to_frame().T)
```



created_utc	2026-09-08 00:00:00+00:00	2026-09-09 00:00:00+00:00	2026-09-10 00:00:00+00:00	2026-09-11 00:00:00+00:00	2026-09-12 00:00:00+00:00	2026-09-13 00:00:00+00:00	2026-09-14 00:00:00+00:00	2026-09-15 00:00:00+00:00	2026-09-16 00:00:00+00:00
posts per day	10	52	28	17	11	16	158	289	1

3. Applying the Round 2 NLP model

The saved Round 2 artefact is loaded exactly as trained: a weighted ensemble of TF-IDF logistic regression, MiniLM sentence embeddings + logistic regression, and three fine-tuned MiniLM seeds.

One compatibility fix is needed: the pickles were written by a newer scikit-learn than the one installed here, which no longer sets `multi_class` on `LogisticRegression`. The attribute is restored to the value the model was trained with (`multinomial`) rather than retraining, so the Round 2 model is used unchanged.

Scoring configuration: truncation and the neutral bias

Two details of the saved Round 2 model matter here, and both are handled explicitly.

Truncation. The fine-tuned MiniLM component was trained on tweets with `max_len = 50` tokens. The posts collected here are longer (median 74 tokens; 68% exceed 50), so a plain call would score only the opening of two thirds of the corpus. Long posts are therefore split into overlapping word windows, scored separately, and their probabilities averaged. The TF-IDF and embedding components see the full text as before.

Neutral bias. The Round 2 artefact carries a validation-selected bias of 1.05 on the Neutral class. The saved `predict()` applies it, while `predict_proba()` does not, so calling `predict_proba` and taking the argmax would silently use a different operating point than Round 2 chose. The bias is applied here.

Neither choice is assumed to help: section 3.2 measures both against 150 hand-labelled in-domain posts.

```
In [5]: from sklearn.linear_model import LogisticRegression

def patch_sklearn_compat(obj):
    cand = []
    if hasattr(obj, "steps"):
        cand.append(obj.steps[-1][1])
    elif isinstance(obj, tuple):
        cand.append(obj[1])
    elif isinstance(obj, LogisticRegression):
        cand.append(obj)
    for c in cand:
        if isinstance(c, LogisticRegression) and not hasattr(c, "multi_class"):
            c.multi_class = "multinomial"

engine = nu.SentimentEnsemble(R2 / "model")
for comp in engine.cfg["components"]:
    patch_sklearn_compat(engine._load(comp))
print(json.dumps({k: engine.cfg[k] for k in ["selected_model", "validation_macro_f1", "test_macro_f1"]},
    indent=2))
print("components:", [(c["name"], c["weight"]) for c in engine.cfg["components"]], "| neutral bias:",
    engine.neutral_bias)

def chunk_text(text, words_per_chunk=34, stride=28, max_chunks=8):
    w = str(text).split()
    if len(w) <= words_per_chunk:
        return [str(text)]
    chunks = [" ".join(w[i:i + words_per_chunk]) for i in range(0, len(w), stride)]
    return chunks[:max_chunks] or [str(text)]

def score(texts, chunked=True, bias=True, batch=512):
    # Reproduces the Round 2 ensemble; optionally averages transformer probabilities over chunks
    texts = [str(t) for t in texts]
    lower, cased = nu.clean_series(texts), nu.clean_series_cased(texts)
    total, wsum = np.zeros((len(texts), len(engine.labels))), 0.0
```

```

for comp in engine.cfg["components"]:
    w = comp["weight"]
    if w <= 0:
        continue
    obj = engine._load(comp)
    if comp["kind"] == "sklearn":
        p = np.vstack([obj.predict_proba(texts[i:i + batch]) for i in range(0, len(texts), batch)])
    elif comp["kind"] == "embedding":
        st, clf = obj
        p = clf.predict_proba(st.encode(lower, batch_size=256, normalize_embeddings=True,
show_progress_bar=False))
    else:
        src = cased if comp["cased"] else lower
        if not chunked:
            p = np.mean([nu.transformer_proba(m, tk, src, comp["max_len"]) for m, tk in obj], axis=0)
        else:
            pieces, owner = [], []
            for idx, t in enumerate(src):
                for ch in chunk_text(t):
                    pieces.append(ch)
                    owner.append(idx)
            owner = np.array(owner)
            flat = np.mean([nu.transformer_proba(m, tk, pieces, comp["max_len"]) for m, tk in obj],
axis=0)
            p = np.vstack([flat[owner == i].mean(axis=0) for i in range(len(texts))])
        total += w * p
        wsum += w
    proba = total / wsum
return engine._apply_neutral_bias(proba) if bias else proba

```

``torch_dtype` is deprecated! Use `dtype` instead!`

```

{
    "selected_model": "Weighted ensemble",
    "validation_macro_f1": 0.7113,
    "test_macro_f1": 0.7227
}
components: [('tfidf_logreg', 0.25), ('embedding_logreg', 0.2), ('minilm_finetuned', 0.55)] | neutral bias:
1.05

```

```

In [6]: if SCORED.exists() and not RESCORE and len(pd.read_csv(SCORED)) == len(work):
        scored = pd.read_csv(SCORED)
        scored["created_utc"] = pd.to_datetime(scored.created_utc, format="mixed", utc=True)
        print("loaded cached scores")
    else:
        t0 = time.perf_counter()
        probs = score(work.text_full.tolist(), chunked=True, bias=True)
        scored = work.copy()
        for i, lab in enumerate(engine.labels):
            scored[f"p_{lab.lower()}"] = probs[:, i].round(4)
        scored["sentiment"] = np.array(engine.labels)[probs.argmax(1)]
        scored["confidence"] = probs.max(1).round(4)
        scored.to_csv(SCORED, index=False)
        print(f"scored {len(scored):,} posts in {time.perf_counter()-t0:.0f}s "
              f"({1000*(time.perf_counter()-t0)/len(scored):.1f} ms/post)")
        scored["sentiment_score"] = scored.p_positive - scored.p_negative
        dist = scored.sentiment.value_counts(normalize=True).reindex(engine.labels)
        METRICS["sentiment_distribution"] = dist.round(4).to_dict()
        METRICS["confidence"] = {"mean": round(float(scored.confidence.mean()), 4),
                                "share_above_0.7": round(float((scored.confidence > 0.7).mean()), 4)}
        print(dist.round(3).to_string()); print(METRICS["confidence"])

```

loaded cached scores

sentiment

Negative 0.401

Neutral 0.409

Positive 0.190

{'mean': 0.5695, 'share_above_0.7': 0.149}

3.2 How accurate is the model on *this* data?

Round 2 reported macro-F1 0.723, but that was on labelled tweets. This section measures the model where it is actually used. 150 posts were drawn at random (not stratified by prediction, which would bias the estimate) and labelled by hand against a fixed rubric: **Positive** = approval, praise or a favourable experience; **Negative** = complaint, bug report, criticism or frustration; **Neutral** = factual news, an announcement, or a question with no clear valence. All three scoring configurations are compared on the same posts.

```
In [7]: val = pd.read_csv(OUT / "Entropy_round3_validation_sample.csv")
val = val[val.human_label.notna() & (val.human_label.astype(str) != "")]
configs = {"as-is (truncated, no bias)": dict(chunked=False, bias=False),
          "neutral bias only": dict(chunked=False, bias=True),
          "chunked + neutral bias (used)": dict(chunked=True, bias=True)}
rows, preds = [], {}
for name, kw in configs.items():
    p = score(val.text_full.tolist(), **kw)
    pred = np.array(engine.labels)[p.argmax(1)]
    preds[name] = pred
    rows.append({"configuration": name, "accuracy": accuracy_score(val.human_label, pred),
                "macro_f1": f1_score(val.human_label, pred, average="macro"),
                **{f"F1 {l}": v for l, v in zip(engine.labels,
                f1_score(val.human_label, pred, labels=engine.labels, average=None,
zero_division=0))}}})
val_res = pd.DataFrame(rows).set_index("configuration")
best = val_res.macro_f1.idxmax()

rng = np.random.default_rng(SEED)
hl, bp = val.human_label.to_numpy(), preds[best]
boot = []
for _ in range(N_BOOT):
    i = rng.integers(0, len(val), len(val))
    boot.append(f1_score(hl[i], bp[i], average="macro", zero_division=0))
ci = np.percentile(boot, [2.5, 97.5])
METRICS["in_domain_validation"] = {
    "n": int(len(val)), "human_label_distribution": val.human_label.value_counts().to_dict(),
    "configurations": val_res.round(4).to_dict("index"), "chosen": best,
    "macro_f1_ci95": [round(float(x), 4) for x in ci],
    "round2_test_macro_f1": engine.cfg["test_macro_f1"],
    "confusion_matrix": confusion_matrix(val.human_label, bp, labels=engine.labels).tolist(),
    "per_class": classification_report(val.human_label, bp, labels=engine.labels, output_dict=True,
zero_division=0),
}
display(val_res.round(3))
print(f"chosen: {best} | macro-F1 95% CI [{ci[0]:.3f}, {ci[1]:.3f}] vs {engine.cfg['test_macro_f1']} on
Round 2 tweets")
print(classification_report(val.human_label, bp, labels=engine.labels, digits=3, zero_division=0))
display(pd.DataFrame(confusion_matrix(val.human_label, bp, labels=engine.labels),
                    index=[f"true {l}" for l in engine.labels], columns=[f"pred {l}" for l in
engine.labels]))
```

	accuracy	macro_f1	F1 Negative	F1 Neutral	F1 Positive
configuration					
as-is (truncated, no bias)	0.653	0.632	0.752	0.600	0.545
neutral bias only	0.673	0.653	0.758	0.634	0.566
chunked + neutral bias (used)	0.687	0.666	0.752	0.656	0.591

chosen: chunked + neutral bias (used) | macro-F1 95% CI [0.578, 0.745] vs 0.7227 on Round 2 tweets

	precision	recall	f1-score	support
Negative	0.855	0.671	0.752	70
Neutral	0.652	0.662	0.656	65
Positive	0.448	0.867	0.591	15
accuracy			0.687	150
macro avg	0.651	0.733	0.666	150
weighted avg	0.726	0.687	0.695	150

	pred Negative	pred Neutral	pred Positive
true Negative	47	21	2
true Neutral	8	43	14
true Positive	0	2	13

```
In [8]: # How much does chunking change the corpus-level picture?
sample_n = min(400, len(work))
sub = work.sample(sample_n, random_state=SEED)
p_trunc = score(sub.text_full.tolist(), chunked=False, bias=True)
p_chunk = score(sub.text_full.tolist(), chunked=True, bias=True)
lab_t = np.array(engine.labels)[p_trunc.argmax(1)]
lab_c = np.array(engine.labels)[p_chunk.argmax(1)]
changed = float((lab_t != lab_c).mean())
net = lambda lab: (lab == "Positive").mean() - (lab == "Negative").mean()
METRICS["chunking_effect"] = {"n_sample": int(sample_n), "labels_changed_share": round(changed, 4),
                              "net_truncated": round(float(net(lab_t)), 4), "net_chunked":
round(float(net(lab_c)), 4)}
print(f"On {sample_n} posts, chunking changes {changed:.1%} of labels; "
      f"net sentiment {net(lab_t):+.3f} (truncated) vs {net(lab_c):+.3f} (chunked)")
```

On 400 posts, chunking changes 12.0% of labels; net sentiment -0.185 (truncated) vs -0.198 (chunked)

```
In [9]: display(scored[["created_utc", "source", "sentiment", "confidence", "text_full"]].head(6))
```

	created_utc	source	sentiment	confidence	text_full
0	2026-09-08 03:48:27+00:00	reddit	Negative	0.5295	Suddenly videos stored on my phone won't play ...
1	2026-09-08 05:32:08+00:00	mastodon	Positive	0.5045	One UI 9.0 Beta 2 is now live for the Galaxy S...
2	2026-09-08 08:39:22+00:00	mastodon	Positive	0.5961	Sony Xperia 1 VIII Android 17 Update: What's N...
3	2026-09-08 08:39:22.135000+00:00	mastodon	Positive	0.5961	Sony Xperia 1 VIII Android 17 Update: What's N...
4	2026-09-08 10:53:05.049000+00:00	mastodon	Neutral	0.6490	Sony hat das Update auf Android 17 für einige ...
5	2026-09-08 12:20:07+00:00	reddit	Neutral	0.5007	Camera app in iOS 27 reportedly includes four ...

4. Sentiment analysis and sentiment shifts

Opinion is tracked with a **net sentiment index** = share(Positive) - share(Negative) inside each 6-hour bin. A shift is only reported when the change between two consecutive bins is statistically significant: a two-proportion z-test on the positive share and on the negative share, with Holm correction across all bin boundaries tested.

```
In [10]: BIN = "12h"
g = scored.set_index("created_utc").resample(BIN)
bins = pd.DataFrame({"n": g.size(),
                     "pos": g.apply(lambda d: (d.sentiment == "Positive").sum() if len(d) else 0),
                     "neg": g.apply(lambda d: (d.sentiment == "Negative").sum() if len(d) else 0),
                     "neu": g.apply(lambda d: (d.sentiment == "Neutral").sum() if len(d) else 0)})
bins = bins[bins.n >= 40]
bins["pos_share"] = bins.pos / bins.n
bins["neg_share"] = bins.neg / bins.n
bins["net"] = bins.pos_share - bins.neg_share

def two_prop(c1, n1, c2, n2):
    p = (c1 + c2) / (n1 + n2)
    se = np.sqrt(p * (1 - p) * (1 / n1 + 1 / n2))
    if se == 0:
        return 0.0, 1.0
    z = (c2 / n2 - c1 / n1) / se
    return z, 2 * (1 - stats.norm.cdf(abs(z)))

tests = []
idx = bins.index
for a, b in zip(idx[:-1], idx[1:]):
    r1, r2 = bins.loc[a], bins.loc[b]
    z_pos, p_pos = two_prop(r1.pos, r1.n, r2.pos, r2.n)
    z_neg, p_neg = two_prop(r1.neg, r1.n, r2.neg, r2.n)
```

```

tests.append({"from": a, "to": b, "n_before": int(r1.n), "n_after": int(r2.n),
             "net_before": r1.net, "net_after": r2.net, "delta_net": r2.net - r1.net,
             "p_pos": p_pos, "p_neg": p_neg, "p_min": min(p_pos, p_neg)})
shifts = pd.DataFrame(tests)
m = len(shifts)
order = shifts.p_min.rank(method="first").astype(int)
shifts["p_holm"] = np.minimum(1, shifts.p_min * (m - order + 1))
shifts = shifts.sort_values("p_holm")
SIG = shifts[shifts.p_holm < 0.05]
METRICS["sentiment_shifts"] = shifts.head(6).assign(**{c: shifts[c].astype(str) for c in ["from",
"to"]}).round(4).to_dict("records")
print(f"{len(SIG)} significant shifts (Holm-corrected p < 0.05) out of {m} consecutive-bin boundaries")
display(shifts.head(6).round(4))

# Pre-specified windows: only three comparisons, so the correction is far less punishing
# than scanning every boundary, and each one answers a question set before seeing the data.
L = LAUNCH["iOS 27"]
periods = {
    "Pre-launch (8-14 Sep)": (scored.created_utc < L),
    "Launch +0-24h": (scored.created_utc >= L) & (scored.created_utc < L + pd.Timedelta("24h")),
    "Settling +24-72h": (scored.created_utc >= L + pd.Timedelta("24h")) & (scored.created_utc < L +
pd.Timedelta("72h")),
    "Later (72h+)": (scored.created_utc >= L + pd.Timedelta("72h")),
}
per = pd.DataFrame([{"period": k, "n": int(v.sum()),
                    "pos": (scored.sentiment[v] == "Positive").mean(),
                    "neg": (scored.sentiment[v] == "Negative").mean()} for k, v in periods.items()])
per["net"] = per.pos - per.neg
prespec = []
for a, b in [(0, 1), (1, 2), (2, 3)]:
    ra, rb = per.iloc[a], per.iloc[b]
    z_pos, p_pos = two_prop(round(ra.pos * ra.n), ra.n, round(rb.pos * rb.n), rb.n)
    z_neg, p_neg = two_prop(round(ra.neg * ra.n), ra.n, round(rb.neg * rb.n), rb.n)
    prespec.append({"comparison": f"{ra.period} -> {rb.period}", "n_before": int(ra.n), "n_after":
int(rb.n),
                  "net_before": ra.net, "net_after": rb.net, "delta_net": rb.net - ra.net,
                  "p_pos": p_pos, "p_neg": p_neg, "p_min": min(p_pos, p_neg)})
prespec = pd.DataFrame(prespec)
rank = prespec.p_min.rank(method="first").astype(int)
prespec["p_holm"] = np.minimum(1, prespec.p_min * (len(prespec) - rank + 1))
METRICS["period_sentiment"] = per.round(4).to_dict("records")
METRICS["prespecified_shifts"] = prespec.round(5).to_dict("records")
display(per.round(3)); display(prespec.round(4))

```

1 significant shifts (Holm-corrected p < 0.05) out of 12 consecutive-bin boundaries

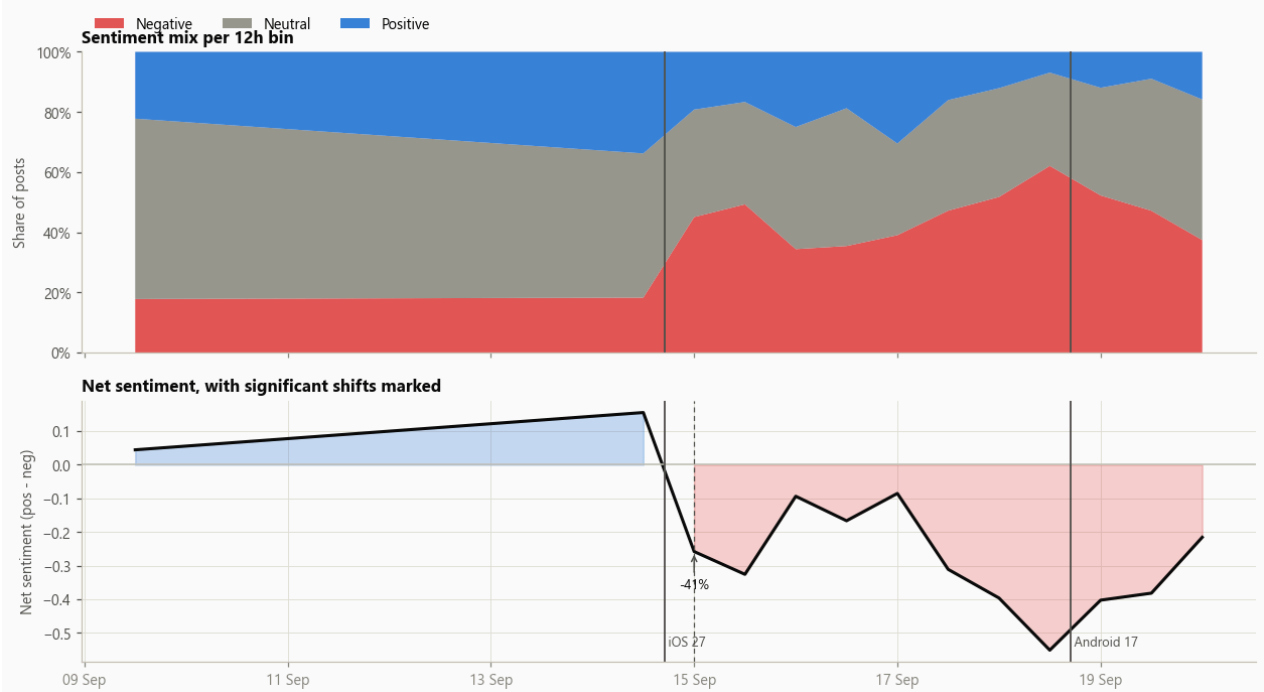
	from	to	n_before	n_after	net_before	net_after	delta_net	p_pos	p_neg	p_min	p_holm
1	2026-09-14 12:00:00+00:00	2026-09-15 00:00:00+00:00	148	151	0.1554	-0.2583	-0.4137	0.0043	0.0000	0.0000	0.0000
6	2026-09-17 00:00:00+00:00	2026-09-17 12:00:00+00:00	82	106	-0.0854	-0.3113	-0.2260	0.0183	0.2641	0.0183	0.2016
3	2026-09-15 12:00:00+00:00	2026-09-16 00:00:00+00:00	138	64	-0.3261	-0.0938	0.2323	0.1627	0.0474	0.0474	0.4744
10	2026-09-19 00:00:00+00:00	2026-09-19 12:00:00+00:00	67	123	-0.4030	-0.3821	0.0209	0.5106	0.5030	0.5030	0.5030
5	2026-09-16 12:00:00+00:00	2026-09-17 00:00:00+00:00	96	82	-0.1667	-0.0854	0.0813	0.0682	0.6194	0.0682	0.6139
11	2026-09-19 12:00:00+00:00	2026-09-20 00:00:00+00:00	123	139	-0.3821	-0.2158	0.1663	0.0937	0.1107	0.0937	0.7498

	period	n	pos	neg	net
0	Pre-launch (8-14 Sep)	171	0.251	0.199	0.053
1	Launch +0-24h	330	0.245	0.367	-0.121
2	Settling +24-72h	355	0.228	0.397	-0.169
3	Later (72h+)	547	0.113	0.486	-0.373

	comparison	n_before	n_after	net_before	net_after	delta_net	p_pos	p_neg	p_min	p_holm
0	Pre-launch (8-14 Sep) -> Launch +0-24h	171	330	0.0526	-0.1212	-0.1738	0.8826	0.0001	0.0001	0.0002
1	Launch +0-24h -> Settling +24-72h	330	355	-0.1212	-0.1690	-0.0478	0.5947	0.4116	0.4116	0.4116
2	Settling +24-72h -> Later (72h+)	355	547	-0.1690	-0.3729	-0.2039	0.0000	0.0086	0.0000	0.0000

```
In [11]: fig, (a1, a2) = plt.subplots(2, 1, figsize=(11.5, 6.4), sharex=True, gridspec_kw={"height_ratios": [1.15, 1]})
share = bins[["neg", "neu", "pos"]].div(bins.n, axis=0)
a1.stackplot(bins.index, share.neg, share.neu, share.pos,
             colors=[SENT["Negative"], SENT["Neutral"], SENT["Positive"]],
             labels=["Negative", "Neutral", "Positive"], alpha=0.92)
a1.set_ylim(0, 1); a1.yaxis.set_major_formatter(PercentFormatter(1.0, decimals=0))
a1.set_ylabel("Share of posts"); a1.set_title(f"Sentiment mix per {BIN} bin")
a1.legend(ncol=3, loc="lower left", bbox_to_anchor=(0, 1.02)); a1.grid(False)

a2.plot(bins.index, bins.net, color=INK, lw=2)
a2.fill_between(bins.index, 0, bins.net, where=bins.net >= 0, color=SENT["Positive"], alpha=0.25)
a2.fill_between(bins.index, 0, bins.net, where=bins.net < 0, color=SENT["Negative"], alpha=0.25)
a2.axhline(0, color=BASE, lw=1)
for _, r in SIG.head(4).iterrows():
    a2.annotate(f"{r.delta_net:+.0%}", xy=(r["to"], bins.loc[r["to"], "net"]), xytext=(0, 18 if r.delta_net > 0 else -24),
               textcoords="offset points", ha="center", fontsize=8.5, color=INK,
               arrowprops=dict(arrowstyle=">", color=INK2, lw=1))
    a2.axvline(r["to"], color=INK2, lw=0.8, ls=(0, (4, 3)))
for label, ts in LAUNCH.items():
    if bins.index.min() <= ts <= bins.index.max():
        for ax in (a1, a2):
            ax.axvline(ts, color=INK2, lw=1.2)
        a2.text(ts, a2.get_ylim()[0] * 0.92, f" {label}", fontsize=8.5, color=INK2)
a2.xaxis.set_major_formatter(mdates.DateFormatter("%d %b"))
a2.set_ylabel("Net sentiment (pos - neg)"); a2.set_title("Net sentiment, with significant shifts marked")
fig.tight_layout()
save(fig, "sentiment_timeline")
```



Is the decline real, or just a change in which platforms we sampled?

Reddit is the most negative platform here and its share of the data grows over the window, so the overall decline could in principle be a sampling artefact. The check below holds the platform fixed.

```
In [12]: comp = pd.crosstab(pd.cut(scored.created_utc,
                                   [scored.created_utc.min() - pd.Timedelta("1h"), L, L + pd.Timedelta("24h"),
                                   L + pd.Timedelta("72h"), scored.created_utc.max() + pd.Timedelta("1h")],
                                   labels=list(periods)), scored.source, normalize="index")
scored["period"] = pd.cut(scored.created_utc,
                           [scored.created_utc.min() - pd.Timedelta("1h"), L, L + pd.Timedelta("24h"),
```

```

L + pd.Timedelta("72h"), scored.created_utc.max() + pd.Timedelta("1h"]],
labels=list(periods))
net = lambda g: (g.sentiment == "Positive").mean() - (g.sentiment == "Negative").mean()
within = (scored.groupby(["source", "period"], observed=True)
        .apply(lambda g: pd.Series({"n": len(g), "net": net(g)}, include_groups=False)
        .reset_index().pivot(index="source", columns="period", values=["n", "net"]))
print("Platform mix per period (share of posts):"); display(comp.round(3))
print("Net sentiment within each platform:"); display(within.round(3))

# Formal check on the platform with good coverage in both the first and last period
mas = scored[scored.source == "mastodon"]
pre_m, late_m = mas[mas.period == "Pre-launch (8-14 Sep)"], mas[mas.period == "Later (72h+)"]
z_p, p_p = two_prop((pre_m.sentiment == "Positive").sum(), len(pre_m), (late_m.sentiment ==
"Positive").sum(), len(late_m))
z_n, p_n = two_prop((pre_m.sentiment == "Negative").sum(), len(pre_m), (late_m.sentiment ==
"Negative").sum(), len(late_m))
METRICS["composition"] = {"platform_mix": comp.round(4).to_dict(),
                          "within_source_net": within["net"].round(4).to_dict(),
                          "mastodon_only": {"n_pre": int(len(pre_m)), "n_late": int(len(late_m)),
                                             "net_pre": round(float(net(pre_m)), 4), "net_late":
round(float(net(late_m)), 4),
                          "positive_z": round(float(z_p), 3), "positive_p": float(p_p),
                          "negative_z": round(float(z_n), 3), "negative_p": float(p_n)}}
print(f"Mastodon only: net {net(pre_m):+.2f} (n={len(pre_m)}) -> {net(late_m):+.2f} (n={len(late_m)}); "
      f"positive share p = {p_p:.2g}, negative share p = {p_n:.2g}")

```

Platform mix per period (share of posts):

	source	hackernews	lemmy	mastodon	reddit
created_utc					
Pre-launch (8-14 Sep)					
		0.076	0.076	0.409	0.439
Launch +0-24h					
		0.276	0.009	0.415	0.300
Settling +24-72h					
		0.073	0.020	0.451	0.456
Later (72h+)					
		0.015	0.020	0.267	0.698

Net sentiment within each platform:

	n				net			
period	Pre-launch (8-14 Sep)	Launch +0-24h	Settling +24-72h	Later (72h+)	Pre-launch (8-14 Sep)	Launch +0-24h	Settling +24-72h	Later (72h+)
source								
hackernews	13.0	91.0	26.0	8.0	-0.154	-0.352	-0.269	-0.375
lemmy	13.0	3.0	7.0	11.0	-0.154	-0.333	-0.286	-0.545
mastodon	70.0	137.0	160.0	146.0	0.429	0.146	0.175	-0.171
reddit	75.0	99.0	162.0	382.0	-0.227	-0.273	-0.488	-0.445

Mastodon only: net +0.43 (n=70) -> -0.17 (n=146); positive share p = 0.00021, negative share p = 3.6e-08

5. Activity analysis and engagement spikes

Two different things can spike: **how much** people post, and **how hard the platform reacts** to what they post. Volume spikes use a robust z-score of hourly counts against a trailing 24-hour median and MAD. Engagement spikes use the same detector on total engagement per bin (Hacker News points, Mastodon favourites + boosts, Lemmy score; Reddit's RSS feed does not expose scores).

```

In [13]: # Robust z-score against a trailing baseline. When the median absolute deviation is zero
# (a long run of quiet hours), fall back to a trailing standard deviation.
def robust_z(series, window=48, min_periods=12):
    med = series.rolling(window, min_periods=min_periods).median().shift(1)
    mad = (series - med).abs().rolling(window, min_periods=min_periods).median().shift(1)
    scale = 1.4826 * mad
    fallback = series.rolling(window, min_periods=min_periods).std().shift(1)
    scale = scale.where(scale > 0, fallback)
    scale = scale.where(scale > 0, np.nan)
    return (series - med) / scale

```

```

vol = scored.set_index("created_utc").resample("1h").size().rename("posts").to_frame()
eng_src = scored[scored.source.isin(["hackernews", "mastodon", "lemmy"])]
vol["engagement"] =
eng_src.set_index("created_utc").score.resample("1h").sum().reindex(vol.index).fillna(0)
vol["z_posts"] = robust_z(vol.posts)
vol["z_engagement"] = robust_z(vol.engagement)
spikes = vol[((vol.z_posts > 3) & (vol.posts >= 6)) | ((vol.z_engagement > 3) & (vol.engagement >=
30))].copy()
spikes["type"] = np.where(spikes.z_posts > 3, np.where(spikes.z_engagement > 3, "volume + engagement",
"volume"), "engagement")
METRICS["spikes"] = [{"hour": str(i), "posts": int(r.posts), "engagement": float(r.engagement),
                      "z_posts": round(float(r.z_posts), 2) if pd.notna(r.z_posts) else None,
                      "z_engagement": round(float(r.z_engagement), 2) if pd.notna(r.z_engagement) else
None,
                      "type": r.type} for i, r in spikes.iterrows()]
print(f"{len(spikes)} spike hours detected (robust z > 3)")
display(spikes[["posts", "engagement", "z_posts", "z_engagement", "type"]].round(2))

```

44 spike hours detected (robust z > 3)

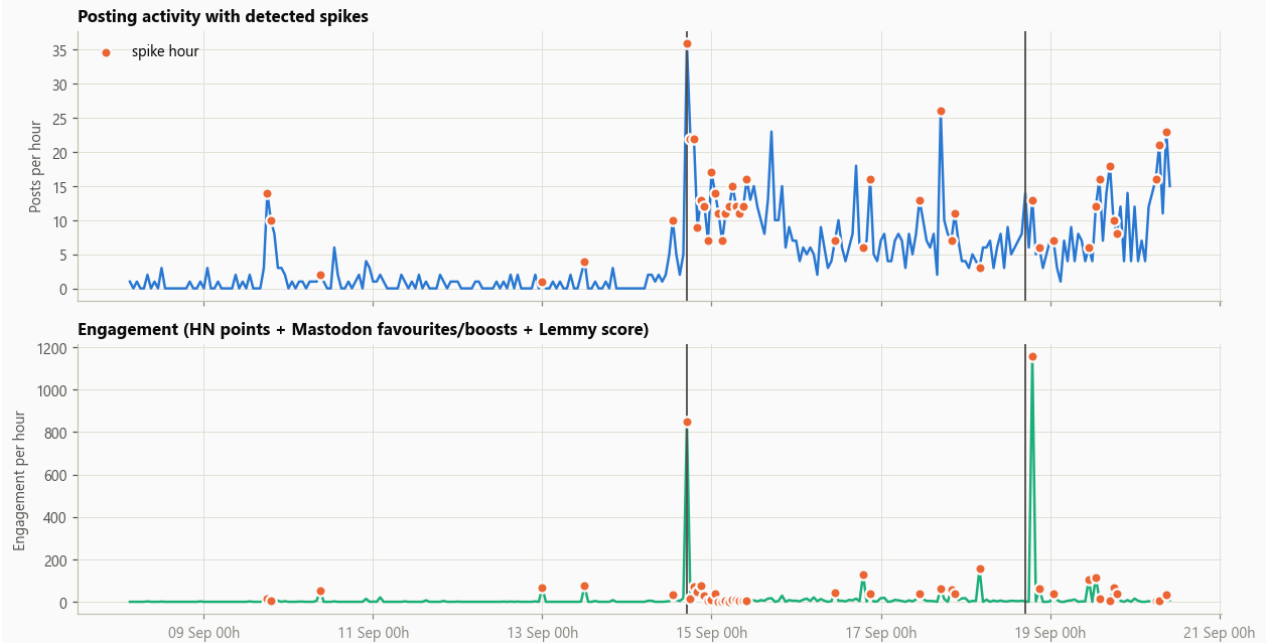
	posts	engagement	z_posts	z_engagement	type
created_utc					
2026-09-09 18:00:00+00:00	14	16.0	14.89	13.07	volume + engagement
2026-09-09 19:00:00+00:00	10	6.0	4.31	2.18	volume
2026-09-10 09:00:00+00:00	2	53.0	0.74	17.69	engagement
2026-09-13 00:00:00+00:00	1	68.0	0.67	21.19	engagement
2026-09-13 12:00:00+00:00	4	76.0	3.60	7.73	volume + engagement
2026-09-14 13:00:00+00:00	10	33.0	26.98	2.27	volume
2026-09-14 17:00:00+00:00	36	851.0	97.13	56.03	volume + engagement
2026-09-14 18:00:00+00:00	22	14.0	59.36	0.11	volume
2026-09-14 19:00:00+00:00	22	71.0	59.36	0.58	volume
2026-09-14 20:00:00+00:00	9	46.0	11.47	0.37	volume
2026-09-14 21:00:00+00:00	13	78.0	8.09	0.63	volume
2026-09-14 22:00:00+00:00	12	29.0	7.42	0.24	volume
2026-09-14 23:00:00+00:00	7	4.0	4.05	0.03	volume
2026-09-15 00:00:00+00:00	17	9.0	10.79	0.07	volume
2026-09-15 01:00:00+00:00	14	38.0	8.77	0.31	volume
2026-09-15 02:00:00+00:00	11	0.0	6.74	0.00	volume
2026-09-15 03:00:00+00:00	7	2.0	4.05	0.02	volume
2026-09-15 04:00:00+00:00	11	4.0	4.27	0.03	volume
2026-09-15 05:00:00+00:00	12	2.0	3.37	0.02	volume
2026-09-15 06:00:00+00:00	15	10.0	4.38	12.82	volume + engagement
2026-09-15 07:00:00+00:00	12	11.0	3.37	6.74	volume + engagement
2026-09-15 08:00:00+00:00	11	6.0	3.04	2.02	volume
2026-09-15 09:00:00+00:00	12	6.0	3.37	1.35	volume
2026-09-15 10:00:00+00:00	16	5.0	4.72	0.67	volume
2026-09-16 11:00:00+00:00	7	43.0	-0.31	5.47	engagement
2026-09-16 19:00:00+00:00	6	129.0	-0.39	20.74	engagement
2026-09-16 21:00:00+00:00	16	37.0	0.79	5.31	engagement
2026-09-17 11:00:00+00:00	13	40.0	1.01	5.90	engagement
2026-09-17 17:00:00+00:00	26	63.0	3.42	9.20	volume + engagement
2026-09-17 20:00:00+00:00	7	56.0	0.00	7.64	engagement
2026-09-17 21:00:00+00:00	11	38.0	0.77	4.95	engagement
2026-09-18 04:00:00+00:00	3	156.0	-0.79	21.44	engagement
2026-09-18 19:00:00+00:00	13	1157.0	2.31	182.99	engagement
2026-09-18 21:00:00+00:00	6	62.0	-0.39	9.78	engagement
2026-09-19 01:00:00+00:00	7	37.0	0.22	5.56	engagement
2026-09-19 11:00:00+00:00	6	106.0	0.00	17.20	engagement
2026-09-19 13:00:00+00:00	12	115.0	2.31	18.72	engagement
2026-09-19 14:00:00+00:00	16	13.0	3.37	1.52	volume
2026-09-19 17:00:00+00:00	18	3.0	4.05	-0.25	volume
2026-09-19 18:00:00+00:00	10	65.0	1.35	10.29	engagement
2026-09-19 19:00:00+00:00	8	40.0	0.67	6.07	engagement
2026-09-20 06:00:00+00:00	16	5.0	3.04	0.34	volume
2026-09-20 07:00:00+00:00	21	3.0	4.72	0.00	volume
2026-09-20 09:00:00+00:00	23	35.0	4.80	5.40	volume + engagement

```
In [14]: fig, (a1, a2) = plt.subplots(2, 1, figsize=(11.5, 6), sharex=True)
a1.plot(vol.index, vol.posts, color=S1, lw=1.6)
a1.scatter(spikes.index, spikes.posts, s=45, color=S2, edgecolor=SURF, linewidth=1.5, zorder=3,
label="spike hour")
```

```

a1.set_ylabel("Posts per hour"); a1.set_title("Posting activity with detected spikes");
a1.legend(loc="upper left")
a2.plot(vol.index, vol.engagement, color=SLOTS[2], lw=1.6)
a2.scatter(spikes.index, spikes.engagement, s=45, color=S2, edgecolor=SURF, linewidth=1.5, zorder=3)
a2.set_ylabel("Engagement per hour"); a2.set_title("Engagement (HN points + Mastodon favourites/boosts + Lemmy score)")
for ax in (a1, a2):
    for label, ts in LAUNCH.items():
        if vol.index.min() <= ts <= vol.index.max():
            ax.axvline(ts, color=INK2, lw=1.2)
a2.xaxis.set_major_formatter(mdates.DateFormatter("%d %b %Hh"))
fig.tight_layout()
save(fig, "activity_spikes")

```



```

In [15]: by_src = (scored.groupby("source")
    .agg(posts=("post_id", "size"), authors=("author_hash", "nunique"),
    pos=("sentiment", lambda s: (s == "Positive").mean()),
    neg=("sentiment", lambda s: (s == "Negative").mean()),
    mean_conf=("confidence", "mean")))
by_src["net"] = by_src.pos - by_src.neg
chi = stats.chi2_contingency(pd.crosstab(scored.source, scored.sentiment))
METRICS["by_source"] = by_src.round(4).to_dict("index")
METRICS["source_sentiment_chi2"] = {"chi2": round(float(chi.statistic), 2), "p": float(chi.pvalue), "dof": int(chi.dof)}
print(f"Sentiment differs by platform: chi2 = {chi.statistic:.1f}, p = {chi.pvalue:.2g}")
display(by_src.round(3))

eng = scored[scored.source.isin(["hackernews", "mastodon", "lemmy"]) & scored.score.notna()]
med = eng.groupby("sentiment").score.median().reindex(engine.labels)
kw = stats.kruskal(*[g.score.values for _, g in eng.groupby("sentiment")])
METRICS["engagement_by_sentiment"] = {"median": med.round(2).to_dict(), "kruskal_p": float(kw.pvalue), "n": int(len(eng))}

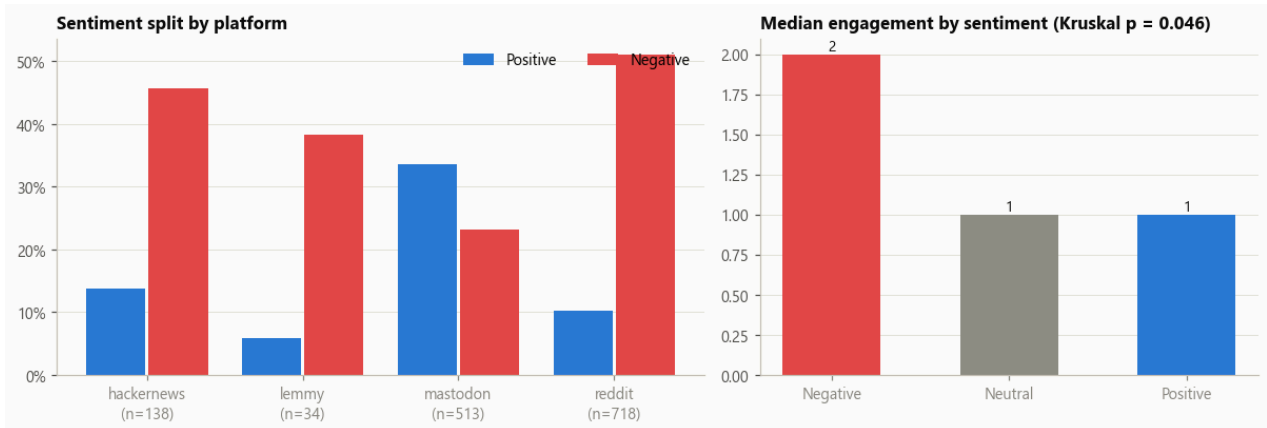
fig, (a1, a2) = plt.subplots(1, 2, figsize=(11, 3.8), gridspec_kw={"width_ratios": [1.3, 1]})
x = np.arange(len(by_src))
a1.bar(x - 0.2, by_src.pos, 0.38, color=SENT["Positive"], label="Positive")
a1.bar(x + 0.2, by_src.neg, 0.38, color=SENT["Negative"], label="Negative")
a1.set_xticks(x, [f"{s}\n(n={int(by_src.loc[s, 'posts'])})" for s in by_src.index])
a1.yaxis.set_major_formatter(PercentFormatter(1.0, decimals=0))
a1.set_title("Sentiment split by platform"); a1.legend(ncol=2, loc="upper right"); a1.grid(axis="x", visible=False)
a2.bar(med.index, med.values, color=[SENT[s] for s in med.index], width=0.55)
for i, v in enumerate(med.values):
    a2.text(i, v, f"{v:.0f}", ha="center", va="bottom", fontsize=9, color=INK)
a2.set_title(f"Median engagement by sentiment (Kruskal p = {kw.pvalue:.3f})"); a2.grid(axis="x", visible=False)

```

```
fig.tight_layout()
save(fig, "platform_engagement")
```

Sentiment differs by platform: $\chi^2 = 153.4$, $p = 1.5e-30$

	posts	authors	pos	neg	mean_conf	net
source						
hackernews	138	120	0.138	0.457	0.565	-0.319
lemmy	34	31	0.059	0.382	0.574	-0.324
mastodon	513	318	0.335	0.232	0.594	0.103
reddit	718	663	0.103	0.511	0.552	-0.408



6. Topic and entity analysis

Three views: which named entities are discussed, which words separate positive from negative reaction (log-odds ratio with an informative Dirichlet prior, Monroe et al. 2008), and unsupervised themes from NMF.

```
In [16]: ent_rows = []
for name in ENTITIES:
    sub = scored[scored[name]]
    if len(sub) >= 25:
        ent_rows.append({"entity": name, "mentions": len(sub), "authors": sub.author_hash.nunique(),
                        "pos": (sub.sentiment == "Positive").mean(), "neg": (sub.sentiment ==
                        "Negative").mean(),
                        "net": (sub.sentiment == "Positive").mean() - (sub.sentiment ==
                        "Negative").mean(),
                        "median_engagement": sub[sub.score.notna()].score.median()})
FEATURES = {"battery": r"\bbattery\b|drain", "performance":
r"\bslow\b|lag\b|performance\b|fast\b|snappy",
            "bugs/crashes": r"\bbug\b|crash\b|broke\b|glitch", "privacy": r"\bprivacy\b|track\b|data
collection",
            "UI/design": r"\bui\b|design\b|icon\b|home screen\b|liquid glass", "battery life after
update": r"battery life",
            "notifications": r"\bnotification", "Siri/assistant": r"\bsiri\b|assistant",
            "install/update process": r"\binstall\b|download\b|update failed\b|rollback\b|downgrade"}
for name, pat in FEATURES.items():
    sub = scored[scored.text_full.str.lower().str.contains(pat, regex=True, na=False)]
    if len(sub) >= 25:
        ent_rows.append({"entity": name, "mentions": len(sub), "authors": sub.author_hash.nunique(),
                        "pos": (sub.sentiment == "Positive").mean(), "neg": (sub.sentiment ==
                        "Negative").mean(),
                        "net": (sub.sentiment == "Positive").mean() - (sub.sentiment ==
                        "Negative").mean(),
                        "median_engagement": sub[sub.score.notna()].score.median()})
ents = pd.DataFrame(ent_rows).sort_values("net")
METRICS["entities"] = ents.round(4).to_dict("records")
display(ents.round(3))

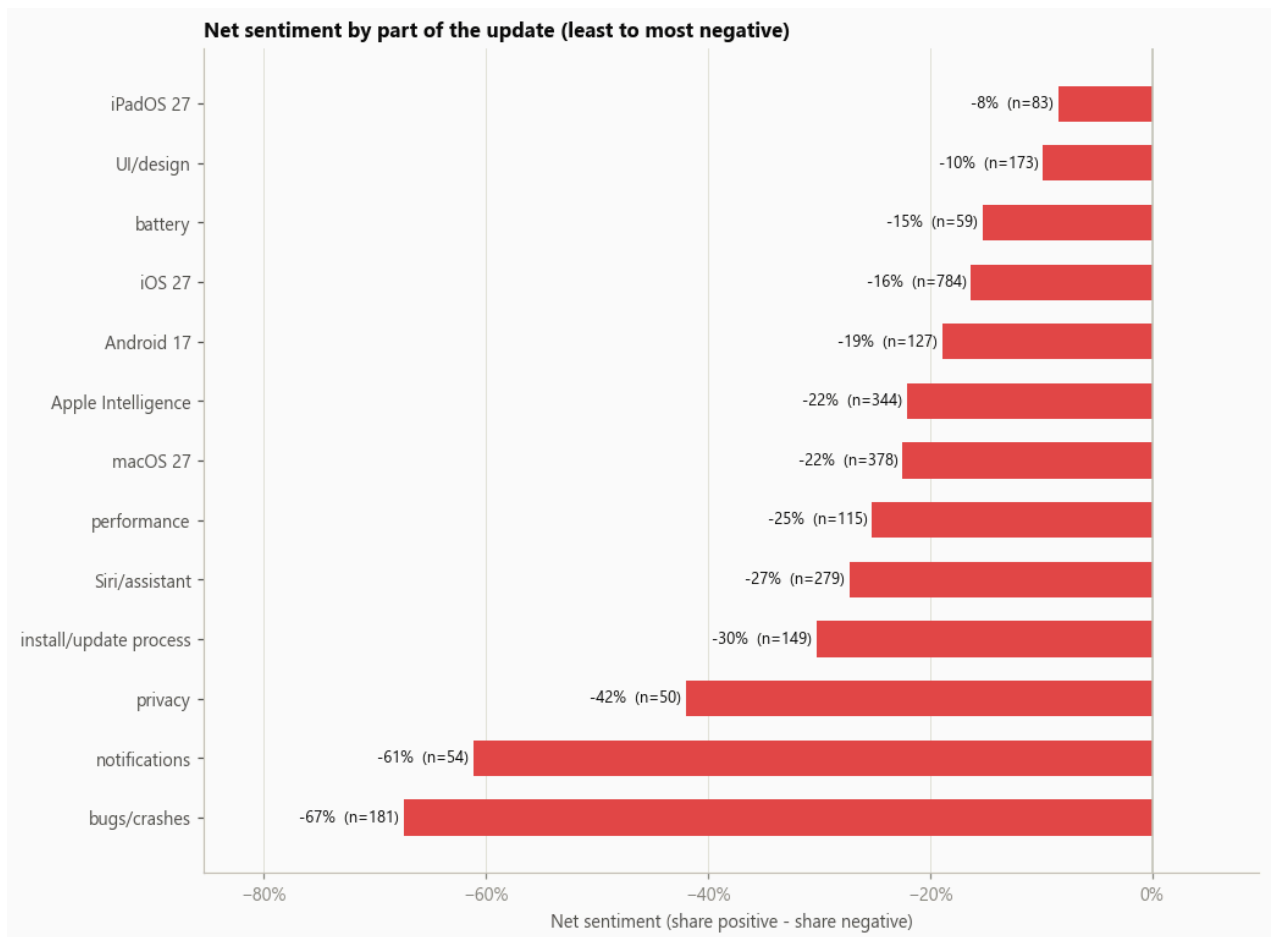
fig, ax = plt.subplots(figsize=(9.5, 0.42 * len(ents) + 1.6))
y = np.arange(len(ents))
```

```

ax.barh(y, ents.net, color=[SENT["Negative"] if v < 0 else SENT["Positive"] for v in ents.net], height=0.6)
for yi, (_, r) in zip(y, ents.iterrows()):
    off = 0.004 if r.net >= 0 else -0.004
    ax.text(r.net + off, yi, f"{r.net:+.0%} (n={int(r.mentions)})", va="center",
            ha="left" if r.net >= 0 else "right", fontsize=8.5, color=INK)
ax.set_yticks(y, ents.entity)
ax.axvline(0, color=BASE, lw=1)
ax.xaxis.set_major_formatter(PercentFormatter(1.0, decimals=0))
ax.set_xlim(ents.net.min() - 0.18, ents.net.max() + 0.18)
ax.set_xlabel("Net sentiment (share positive - share negative)")
ax.set_title("Net sentiment by part of the update (least to most negative)")
ax.grid(axis="y", visible=False)
fig.tight_layout()
save(fig, "entity_sentiment")

```

	entity	mentions	authors	pos	neg	net	median_engagement
7	bugs/crashes	181	169	0.072	0.746	-0.674	1.0
10	notifications	54	48	0.019	0.630	-0.611	7.5
8	privacy	50	50	0.120	0.540	-0.420	2.0
12	install/update process	149	138	0.174	0.477	-0.302	1.0
11	Siri/assistant	279	246	0.129	0.401	-0.272	1.0
6	performance	115	106	0.304	0.557	-0.252	1.0
2	macOS 27	378	318	0.233	0.458	-0.225	1.0
3	Apple Intelligence	344	293	0.157	0.378	-0.221	1.0
4	Android 17	127	104	0.118	0.307	-0.189	1.0
0	iOS 27	784	650	0.207	0.370	-0.163	1.0
5	battery	59	57	0.254	0.407	-0.153	1.0
9	UI/design	173	151	0.254	0.353	-0.098	1.0
1	iPadOS 27	83	68	0.181	0.265	-0.084	1.5



```

In [17]: FEED_STOP = ["submitted", "link", "comments", "https", "http", "com", "www", "amp", "reddit", "post",
"posted"]
STOPWORDS = list(TfidfVectorizer(stop_words="english").get_stop_words()) + FEED_STOP
cv = CountVectorizer(ngram_range=(1, 2), min_df=8, stop_words=STOPWORDS, max_features=8000)
X = cv.fit_transform(scored.text_full.str.lower())
vocab = np.array(cv.get_feature_names_out())
pos_mask = (scored.sentiment == "Positive").values
neg_mask = (scored.sentiment == "Negative").values
y_pos = np.asarray(X[pos_mask].sum(0)).ravel()
y_neg = np.asarray(X[neg_mask].sum(0)).ravel()
alpha = np.asarray(X.sum(0)).ravel()
a0 = alpha.sum()
def logodds(y_i, y_j):
    n_i, n_j = y_i.sum(), y_j.sum()
    d_i = np.log((y_i + alpha) / (n_i + a0 - y_i - alpha))
    d_j = np.log((y_j + alpha) / (n_j + a0 - y_j - alpha))
    var = 1 / (y_i + alpha) + 1 / (y_j + alpha)
    return (d_i - d_j) / np.sqrt(var)
z = logodds(y_pos, y_neg)
top_pos = vocab[np.argsort(-z)[:15]]
top_neg = vocab[np.argsort(z)[:15]]
METRICS["log_odds"] = {"positive": list(top_pos), "negative": list(top_neg)}
display(pd.DataFrame({"most positive-leaning": top_pos, "most negative-leaning": top_neg}))

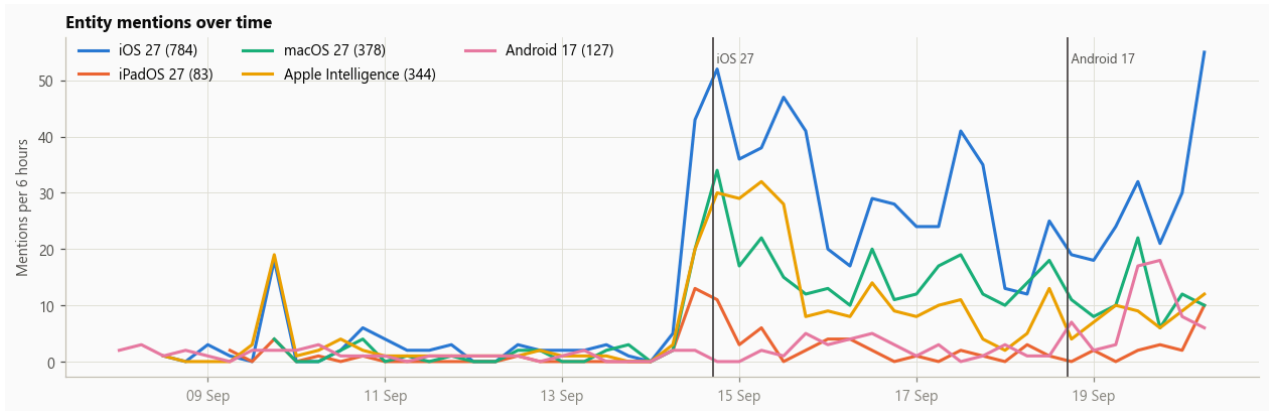
tf = TfidfVectorizer(max_features=6000, min_df=8, stop_words=STOPWORDS, ngram_range=(1, 2))
V = tf.fit_transform(scored.text_full.str.lower())
nmf = NMF(n_components=6, random_state=42, init="nndsvda", max_iter=400).fit(V)
terms = np.array(tf.get_feature_names_out())
W = nmf.transform(V)
scored["nmf_topic"] = W.argmax(1)
topics = []
for k in range(nmf.n_components):
    sub = scored[scored.nmf_topic == k]
    topics.append({"topic": k, "size": len(sub), "share": len(sub) / len(scored),
                  "top_terms": ", ".join(terms[np.argsort(-nmf.components_[k][:8])[:8]]),
                  "net_sentiment": (sub.sentiment == "Positive").mean() - (sub.sentiment ==
"Negative").mean())}
topics = pd.DataFrame(topics).sort_values("size", ascending=False)
METRICS["nmf_topics"] = topics.round(4).to_dict("records")
display(topics.round(3))

```

	most positive-leaning	most negative-leaning
0	new	don
1	ios27	storage
2	2026	issue
3	improvements	fix
4	galaxy	tried
5	health	bug
6	new features	problem
7	apple	mac
8	available	external
9	apple app	working
10	features	mail
11	faster	messages
12	ui	macbook
13	9to5mac	menu bar
14	apps apple	account

	topic	size	share	top_terms	net_sentiment
0	0	392	0.279	ios, ios 27, 27, iphone, 26, ios 26, update, beta	-0.311
1	1	299	0.213	macos, macos 27, 27, golden, golden gate, gate...	-0.284
5	5	233	0.166	ios27, app, macos27, ios27 macos27, new, apple...	0.129
3	3	207	0.148	siri, ai, siri ai, new siri, new, just, use, turn	-0.338
2	2	141	0.100	android, android 17, 17, update, pixel, qpr1, ...	-0.220
4	4	131	0.093	apple, intelligence, apple intelligence, featu...	-0.130

```
In [18]: fig, ax = plt.subplots(figsize=(11.5, 3.8))
ent_time = scored.set_index("created_utc")
for name, c in zip(ENTITIES, SLOTS):
    s = ent_time[ent_time[name]].resample("6h").size()
    if s.sum() >= 30:
        ax.plot(s.index, s.values, color=c, lw=2, label=f"{name} ({int(s.sum())})")
for label, ts in LAUNCH.items():
    if ent_time.index.min() <= ts <= ent_time.index.max():
        ax.axvline(ts, color=INK2, lw=1.2)
        ax.text(ts, ax.get_ylim()[1] * 0.95, f" {label}", fontsize=8.5, color=INK2, va="top")
ax.xaxis.set_major_formatter(mdates.DateFormatter("%d %b"))
ax.set_ylabel("Mentions per 6 hours"); ax.set_title("Entity mentions over time")
ax.legend(ncol=3, loc="upper left", bbox_to_anchor=(0, 1.02))
fig.tight_layout()
save(fig, "entity_timeline")
```



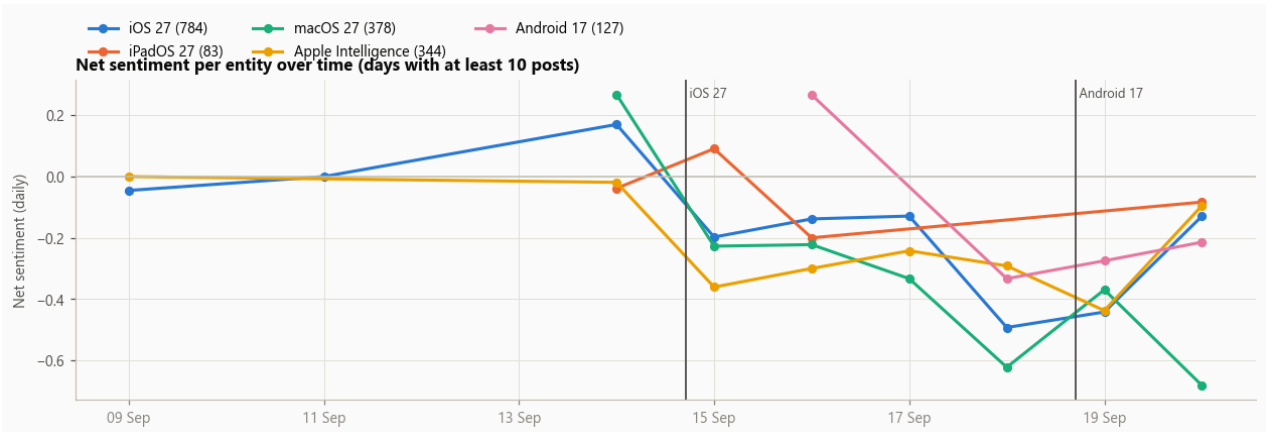
```
In [19]: # Sentiment (not just volume) per entity over time
tracked = [e for e in ENTITIES if scored[e].sum() >= 80]
fig, ax = plt.subplots(figsize=(11.5, 4))
for name, c in zip(tracked, SLOTS):
    g = scored[scored[name]].set_index("created_utc").resample("1D")
    n = g.size()
    net = g.sentiment.apply(lambda x: (x == "Positive").mean() - (x == "Negative").mean() if len(x) >= 10
    else np.nan)
    keep = n >= 10
    ax.plot(net.index[keep], net[keep], color=c, lw=2, marker="o", ms=5, label=f"{name} ({int(scored[name].sum())})")
ax.axhline(0, color=BASE, lw=1)
for label, ts in LAUNCH.items():
    if scored.created_utc.min() <= ts <= scored.created_utc.max():
        ax.axvline(ts, color=INK2, lw=1.2)
        ax.text(ts, ax.get_ylim()[1] * 0.92, f" {label}", fontsize=8.5, color=INK2, va="top")
ax.xaxis.set_major_formatter(mdates.DateFormatter("%d %b"))
ax.set_ylabel("Net sentiment (daily)")
ax.set_title("Net sentiment per entity over time (days with at least 10 posts)")
ax.legend(ncol=3, loc="lower left", bbox_to_anchor=(0, 1.02))
fig.tight_layout()
save(fig, "entity_sentiment_timeline")

ent_daily = {}
for name in tracked:
    g = scored[scored[name]].set_index("created_utc").resample("1D")
    n, net = g.size(), g.sentiment.apply(lambda x: (x == "Positive").mean() - (x == "Negative").mean() if
```

```

len(x) >= 10 else np.nan)
    ent_daily[name] = {str(d.date()): round(float(v), 4) for d, v, c in zip(net.index, net, n) if c >= 10
and pd.notna(v)}
METRICS["entity_daily_net"] = ent_daily
print(json.dumps(ent_daily, indent=1)[:900])

```



```

{
  "iOS 27": {
    "2026-09-09": -0.0455,
    "2026-09-11": 0.0,
    "2026-09-14": 0.17,
    "2026-09-15": -0.1975,
    "2026-09-16": -0.1383,
    "2026-09-17": -0.129,
    "2026-09-18": -0.4928,
    "2026-09-19": -0.4421,
    "2026-09-20": -0.1294
  },
  "iPadOS 27": {
    "2026-09-14": -0.0385,
    "2026-09-15": 0.0909,
    "2026-09-16": -0.2,
    "2026-09-20": -0.0833
  },
  "macOS 27": {
    "2026-09-14": 0.2679,
    "2026-09-15": -0.2273,
    "2026-09-16": -0.2222,
    "2026-09-17": -0.3333,
    "2026-09-18": -0.6226,
    "2026-09-19": -0.3696,
    "2026-09-20": -0.6818
  },
  "Apple Intelligence": {
    "2026-09-09": 0.0,
    "2026-09-14": -0.0189,
    "2026-09-15": -0.3608,
    "2026-09-16": -0.3,
    "2026-09-17": -0.2424,
    "2026-09-18": -0.2917,
    "2026-09-19": -0.4375,
    "2026-09-20": -0.0952
  },
  "Android 17": {
    "2026-09-16": 0.2667,
    "2026-09-18": -0.3333,
    "2026-09-19": -0.275,
    "2026-09-20": -0.2143
  }
}

```

7. Trigger explanations

Each spike and each significant sentiment shift is matched against two independent sources of evidence: news articles published in the same window (Google News RSS) and the most-engaged posts written inside that window. Nothing is inferred from the model alone.

```
In [20]: def window_evidence(ts, hours_before=6, hours_after=6, k=4):
    lo, hi = ts - pd.Timedelta(hours=hours_before), ts + pd.Timedelta(hours=hours_after)
    n = news[(news.published_utc >= lo) & (news.published_utc <= hi)]
    p = scored[(scored.created_utc >= lo) & (scored.created_utc <= hi)]
    top_posts = p.sort_values("score", ascending=False).head(k) if p.score.notna().any() else p.head(k)
    return n.head(k), top_posts

triggers = []
for _, r in spikes.iterrows():
    n, p = window_evidence(r.name)
    triggers.append({"when": str(r.name), "kind": f"{r.type} spike",
                    "size": f"{int(r.posts)} posts/h (z={r.z_posts:.1f})" if pd.notna(r.z_posts) else
                    f"engagement z={r.z_engagement:.1f}",
                    "headlines": list(n.title.head(3)), "top_post": (p.text_full.iloc[0][:180] if len(p)
                    else "")})
for _, r in SIG.head(4).iterrows():
    n, p = window_evidence(r["to"])
    triggers.append({"when": str(r["to"]), "kind": "sentiment shift",
                    "size": f"net {r.net_before:+.0%} -> {r.net_after:+.0%} (Holm p={r.p_holm:.3f})",
                    "headlines": list(n.title.head(3)), "top_post": (p.text_full.iloc[0][:180] if len(p)
                    else "")})
METRICS["triggers"] = triggers
for t in triggers:
    print(f"\n== {t['when']} | {t['kind']} | {t['size']}")
    for h in t["headlines"]:
        print("    news:", h[:120])
    if t["top_post"]:
        print("    post:", t["top_post"][:150])
```

=== 2026-09-09 18:00:00+00:00 | volume + engagement spike | 14 posts/h (z=14.9)
news: Apple ushers in new, AI-powered era for Siri - Mashable
post: Upcoming Vernissage Update for iOS 27 * **Submitted Build**: A new version containing small fixes for iOS 27 was recently submitted to Apple for review

=== 2026-09-09 19:00:00+00:00 | volume spike | 10 posts/h (z=4.3)
news: Apple ushers in new, AI-powered era for Siri - Mashable
post: Upcoming Vernissage Update for iOS 27 * **Submitted Build**: A new version containing small fixes for iOS 27 was recently submitted to Apple for review

=== 2026-09-10 09:00:00+00:00 | engagement spike | 2 posts/h (z=0.7)
news: How to Prepare for iOS 27 on iPhone & iPadOS 27 on iPad - osxdaily.com
news: Apple Intelligence is getting usage limits, and here's what it means for you - digitaltrends.com
news: iOS 27 and macOS 27 has an Easter egg that will delight longtime Apple fans - macworld.com
post: The new Apple Watch can listen in on your conversations and take notes. That makes me feel weird.
::: spoiler article text So you got an Apple Watch t

=== 2026-09-13 00:00:00+00:00 | engagement spike | 1 posts/h (z=0.7)
post: >Apple is relying on an opt-in procedure: When activating Siri AI, the manufacturer asks for help in improving the chatbot, as developers with access

=== 2026-09-13 12:00:00+00:00 | volume + engagement spike | 4 posts/h (z=3.6)
news: iOS 27 breaks 15 years of muscle memory on iPhone and iPad - 9to5Mac
news: Apple Intelligence could be the key to cram Face ID into future iPhone Duo - AppleInsider
news: macOS 27 Golden Gate on M1 and M2 Macs: What Changes Monday, What Doesn't - The Mac Observer
post: Backflip: Apple now wants to train AI models with user data after all >Apple is relying on an opt-in procedure: When activating Siri AI, the manufacturer

=== 2026-09-14 13:00:00+00:00 | volume spike | 10 posts/h (z=27.0)
news: iOS 27 and macOS 27 are loaded with security fixes-here's why - macworld.com
news: I've used macOS Golden Gate for 3 months - and here's what I think - Tom's Guide
news: Apple Intelligence Languages in iOS 27: All 16, Plus What Siri AI Adds in October - The Mac Observer
post: iOS 27, iPadOS 27, and macOS 27

=== 2026-09-14 17:00:00+00:00 | volume + engagement spike | 36 posts/h (z=97.1)
news: iOS 27, Siri AI, and iPadOS 27 compatibility explained - here's which devices are supported - TechRadar
news: When does iOS 27 come out? Date and time you can download the new iPhone operating system around the world - Fast Company
news: How To Prepare Your iPhone For The iOS 27 Update - Engadget
post: iOS 27, iPadOS 27, and macOS 27

=== 2026-09-14 18:00:00+00:00 | volume spike | 22 posts/h (z=59.4)
news: When does iOS 27 come out? Date and time you can download the new iPhone operating system around the world - Fast Company
news: How To Prepare Your iPhone For The iOS 27 Update - Engadget
news: iOS and iPadOS 27: The MacStories Review - macstories.net
post: iOS 27, iPadOS 27, and macOS 27

=== 2026-09-14 19:00:00+00:00 | volume spike | 22 posts/h (z=59.4)
news: How To Prepare Your iPhone For The iOS 27 Update - Engadget
news: iOS and iPadOS 27: The MacStories Review - macstories.net
news: Our OS 27 Review Extras: eBooks, Shortcuts, and the Making of the iOS and iPadOS 27 Review - macstories.net
post: iOS 27, iPadOS 27, and macOS 27

=== 2026-09-14 20:00:00+00:00 | volume spike | 9 posts/h (z=11.5)
news: iOS and iPadOS 27: The MacStories Review - macstories.net
news: Our OS 27 Review Extras: eBooks, Shortcuts, and the Making of the iOS and iPadOS 27 Review - macstories.net
news: macOS 27: The MacStories Review - Page 9 of 15 - macstories.net
post: iOS 27, iPadOS 27, and macOS 27

=== 2026-09-14 21:00:00+00:00 | volume spike | 13 posts/h (z=8.1)
news: iOS and iPadOS 27: The MacStories Review - macstories.net
news: Our OS 27 Review Extras: eBooks, Shortcuts, and the Making of the iOS and iPadOS 27 Review - macstories.net
news: macOS 27: The MacStories Review - Page 9 of 15 - macstories.net
post: iOS 27, iPadOS 27, and macOS 27

=== 2026-09-14 22:00:00+00:00 | volume spike | 12 posts/h (z=7.4)
news: iOS 27 Available Now With These 8 New Features - MacRumors
news: watchOS 27: The MacStories Review - macstories.net
news: 10 Cool New iPhone Features You Need to Try Now - gearpatrol.com
post: iOS 27, iPadOS 27, and macOS 27

=== 2026-09-14 23:00:00+00:00 | volume spike | 7 posts/h (z=4.0)
news: The 12 best iOS 27, iPadOS 27 and macOS 27 tips you might have missed - stuff.tv
news: Fashion app Daydream uses Apple Intelligence to help you shop the outfits in your camera roll - techcrunch.com
news: Apple Releases iOS 27 and iPadOS 27 With Siri AI and Liquid Glass Update - MacRumors
post: iOS 27, iPadOS 27, and macOS 27

=== 2026-09-15 00:00:00+00:00 | volume spike | 17 posts/h (z=10.8)
news: Jamf Announces Same-Day Support For macOS Golden Gate 27, iOS 27 And Other OS 27 Releases - Yahoo Finance
news: Jamf Announces Same-Day Support For macOS Golden Gate 27, iOS 27 And Other OS 27 Releases - Business Wire
news: Apple releases new parental controls with iOS 27, iPadOS 27, and macOS 27 - BetaNews
post: Tell HN: iOS 27 does not allow Apple Intelligence to be disabled I'm surprised that this didn't get more attention during the Beta period.<p

=== 2026-09-15 01:00:00+00:00 | volume spike | 14 posts/h (z=8.8)
news: The next generation of Apple Intelligence is available today - Apple
news: Apple releases test of redesigned Siri AI before iPhone 18 hits stores this week - CNBC
news: Apple Rolls Out iOS 27, iPadOS 27, macOS 27, and More to the Public; Here's What's New - Appleosophy
post: Tell HN: iOS 27 does not allow Apple Intelligence to be disabled I'm surprised that this didn't get more attention during the Beta period.<p

=== 2026-09-15 02:00:00+00:00 | volume spike | 11 posts/h (z=6.7)
news: iOS 27 is out! What Florida iPhone users should check before installing - Florida Today
news: iPadOS 27 brings Siri AI and system optimizations to your iPad - AppleInsider
news: Apple released iOS 27 and iPadOS 27 - what's new and which iPhone and iPad models can be updated - m obidevices.com
post: Tell HN: iOS 27 does not allow Apple Intelligence to be disabled I'm surprised that this didn't get more attention during the Beta period.<p

=== 2026-09-15 03:00:00+00:00 | volume spike | 7 posts/h (z=4.0)
news: Apple Releases macOS 27 Golden Gate With Siri AI and Liquid Glass Refinements: how 26 outlets framed it - NewsCord
news: Here's how to download the macOS Golden Gate update right now - Tom's Guide
news: Here's every security fix included in iOS 27 and iPadOS 27 - 9to5Mac
post: Tell HN: iOS 27 does not allow Apple Intelligence to be disabled I'm surprised that this didn't get more attention during the Beta period.<p

=== 2026-09-15 04:00:00+00:00 | volume spike | 11 posts/h (z=4.3)
news: macOS Golden Gate: 10 Features to Try First - MacRumors
news: Pixel 11 Gets Another September Update, but Android 17 QPR1 Is Still Missing - TechRepublic
news: iOS 27 Features: The Ultimate Mega Guide - MacRumors
post: Apple found a new way to sabotage and ruin PWAs in iOS 27. When you add a PWA to the Home Screen, it forces this fade onto it, and you can't remove it

=== 2026-09-15 05:00:00+00:00 | volume spike | 12 posts/h (z=3.4)
news: macOS Golden Gate: 10 Features to Try First - MacRumors
news: Pixel 11 Gets Another September Update, but Android 17 QPR1 Is Still Missing - TechRepublic
news: iOS 27 Features: The Ultimate Mega Guide - MacRumors
post: Apple found a new way to sabotage and ruin PWAs in iOS 27. When you add a PWA to the Home Screen, it forces this fade onto it, and you can't remove it

=== 2026-09-15 06:00:00+00:00 | volume + engagement spike | 15 posts/h (z=4.4)
news: macOS Golden Gate: 10 Features to Try First - MacRumors
news: Pixel 11 Gets Another September Update, but Android 17 QPR1 Is Still Missing - TechRepublic
news: iOS 27 Features: The Ultimate Mega Guide - MacRumors
post: Apple found a new way to sabotage and ruin PWAs in iOS 27. When you add a PWA to the Home Screen, it forces this fade onto it, and you can't remove it

=== 2026-09-15 07:00:00+00:00 | volume + engagement spike | 12 posts/h (z=3.4)
news: Here's what's new with parental control features in iOS 27, iPadOS 27, and macOS 27 - 9to5Mac
news: Apple overhauls security with iOS and macOS 27 - iNews
news: Apple Releases iOS 27 With Key New iPhone Features And Major Upgrades - Forbes

post: Apple found a new way to sabotage and ruin PWAs in iOS 27. When you add a PWA to the Home Screen, it forces this fade onto it, and you can't remove it

=== 2026-09-15 08:00:00+00:00 | volume spike | 11 posts/h (z=3.0)

news: Here's what's new with parental control features in iOS 27, iPadOS 27, and macOS 27 - 9to5Mac

news: Apple overhauls security with iOS and macOS 27 - iNews

news: Apple Releases iOS 27 With Key New iPhone Features And Major Upgrades - Forbes

post: Kinda surprised to see this. # macOS27

=== 2026-09-15 09:00:00+00:00 | volume spike | 12 posts/h (z=3.4)

news: Apple Releases iOS 27 With Key New iPhone Features And Major Upgrades - Forbes

news: 50 New Things Your iPhone Can Do in iOS 27 - Out Now! - MacRumors

news: macOS 27 Golden Gate: Intel Macs Dropped [2026] - tech-insider.org

post: Kinda surprised to see this. # macOS27

=== 2026-09-15 10:00:00+00:00 | volume spike | 16 posts/h (z=4.7)

news: 50 New Things Your iPhone Can Do in iOS 27 - Out Now! - MacRumors

news: macOS 27 Golden Gate: Intel Macs Dropped [2026] - tech-insider.org

news: iOS 27 Brings These Five New Features to Your iPhone Lock Screen - MacRumors

post: Upgraded from macOS 15 to 27 and fuck me, I hate the bubbles everywhere. The tab bar in Safari is an abomination. All toolbars everywhere inspire disg

=== 2026-09-16 11:00:00+00:00 | engagement spike | 7 posts/h (z=-0.3)

news: Apple fixed over 200 flaws in iOS 27 and macOS 27 - Korben

news: macOS 27 Golden Gate has arrived: Here's what's new on your Mac - macworld.com

news: Why This Single iOS 27 Feature Turns the iPhone Duo Into a Pocket Laptop - PCMag

post: "As it turns out, Apple realistically needs customer data to make Siri AI work well. So, starting with iOS 27, it is asking users to opt-in to trainin

=== 2026-09-16 19:00:00+00:00 | engagement spike | 6 posts/h (z=-0.4)

news: iOS 27: Customize Controls in the Camera App - MacRumors

news: iOS 27 adds three new iPhone features you'll use all the time - 9to5Mac

news: Samsung starts rolling out Android 17 with One UI 9, confirms initial list of Galaxy devices - 9to5Google
post: macOS 27 Golden Gate - Review

=== 2026-09-16 21:00:00+00:00 | engagement spike | 16 posts/h (z=0.8)

news: 7 New iOS 27 Features You'll Want to Try Out - osxdaily.com

news: Apple Weather Adds two convenient new features in iOS 27 - 9to5Mac

news: macOS 27 Golden Gate closes the door on Intel Macs - The Apple Post

post: macOS 27 Golden Gate - Review

=== 2026-09-17 11:00:00+00:00 | engagement spike | 13 posts/h (z=1.0)

news: An iOS 27 Bug Can Temporarily Freeze Your iPhone - Engadget

news: Safari Keeps Getting Better. Try These 21 Hidden Tricks Now That iOS 27 Is Here - PCMag

news: Much-Anticipated Apple Software Rollout Starts: What Users Should Know - DailyVoice.com

post: I have a MacBook Pro M4, and I have Siri totally disabled. I also have a (linked) iPhone 15 with Siri totally disabled. I upgraded the computer to mac

=== 2026-09-17 17:00:00+00:00 | volume + engagement spike | 26 posts/h (z=3.4)

news: Google Home just became a power user's paradise - 9to5Google

news: Which iOS 27 CarPlay Features Actually Work in Your Car? - MacRumors

news: Apple Watch Series 12: Worthwhile upgrade, looking forward to Audio Intelligence - MobileSyrup

post: Android 17 sucks. So I put Linux on a phone.

=== 2026-09-17 20:00:00+00:00 | engagement spike | 7 posts/h (z=0.0)

news: iOS 27 CarPlay Features Compatibility: Cars, iPhones, Apps - Gadget Hacks

news: Another Motorola phone just picked up its big Android 17 update - Android Authority

news: iPadOS 27.2 adds convenient new feature from the Mac - 9to5Mac

post: Android 17 sucks. So I put Linux on a phone.

=== 2026-09-17 21:00:00+00:00 | engagement spike | 11 posts/h (z=0.8)

news: One UI 9: Samsung Reveals When Your Galaxy Will Be Updated With Android 17 - Ubergizmo

news: Larry Magid: iOS 27 brings a smarter Siri and more - The Mercury News

news: Honor confirms list of devices getting its Liquid Glass-filled Android 17 update [Video] - 9to5Google
e
post: Android 17 sucks. So I put Linux on a phone.

=== 2026-09-18 04:00:00+00:00 | engagement spike | 3 posts/h (z=-0.8)

news: Android 17 QPR2 Beta 5: key changes - NeoTeo

news: Apple Watch Series 12 Is Becoming The AI Wearable You Already Own - WeRSM
news: Google Keep update tweaks the homescreen widget on Android - 9to5Google
post: GrapheneOS accuses Google of gatekeeping Android 17 features and security fixes

=== 2026-09-18 19:00:00+00:00 | engagement spike | 13 posts/h (z=2.3)
news: watchOS 27 kills the Digital Crown double-click, and Apple Watch owners aren't happy - Yahoo Tech
news: iPhone 18 Pro: The New Features You Should Try First - MacRumors
news: iOS 27—Apple Just Gave iPhone Users 120 Reasons To Update Now - Forbes
post: Android 17 is the first since 3.x to add new APIs without releasing to the AOSP

=== 2026-09-18 21:00:00+00:00 | engagement spike | 6 posts/h (z=-0.4)
news: Siri AI vs ChatGPT: I gave iOS 27's new assistant 5 jobs I rely on ChatGPT Voice for, and it's changing how I use AI - T
news: Siri AI vs ChatGPT: I gave iOS 27's new assistant 5 jobs I rely on ChatGPT Voice for, and it's changing how I use AI - Y
news: Siri app vs. Hey Siri: how to get more from Siri AI - AppleInsider
post: Android 17 is the first since 3.x to add new APIs without releasing to the AOSP

=== 2026-09-19 01:00:00+00:00 | engagement spike | 7 posts/h (z=0.2)
news: Apple Loop: iPhone 18 Pro Camera, iOS 27 Changes, M6 MacBook Benchmarks - Yahoo Tech
news: iPhone 18 Pro, Apple Watch Series 12 and AirPods 5 hit stores - Yahoo Tech
news: Apple iPhone 13 vs. iPhone 18: How Performance and Features Have Changed in 5 Years - PCMag
post: Android 17 is the first since 3.x to add new APIs without releasing to the AOSP

=== 2026-09-19 11:00:00+00:00 | engagement spike | 6 posts/h (z=0.0)
news: iOS 27: 10 Secret iPhone Features Apple Forgot to Mention! - Geeky Gadgets
news: Notify Me is easily one of the best features iOS 27 has to offer – but it needs to do better - Tom's Guide
news: Notify Me is easily one of the best features iOS 27 has to offer – but it needs to do better - Yahoo Tech
post: Android 17 is the first since 3.x to add new APIs without releasing to the AOSP

=== 2026-09-19 13:00:00+00:00 | engagement spike | 12 posts/h (z=2.3)
news: Notify Me is easily one of the best features iOS 27 has to offer – but it needs to do better - Tom's Guide
news: Notify Me is easily one of the best features iOS 27 has to offer – but it needs to do better - Yahoo Tech
news: Android 17: Samsung rolls out One UI 9.0 to Galaxy S26, 26+, 26 Ultra - Deccan Herald
post: Android 17 is the first since 3.x to add new APIs without releasing to the AOSP

=== 2026-09-19 14:00:00+00:00 | volume spike | 16 posts/h (z=3.4)
news: I replaced ChatGPT with Siri AI for a day – here's what surprised me - Yahoo Tech
news: Apple's \$250 million Siri settlement: who qualifies and how claims will work - The Mac Observer
news: Apple's own pages disagree on when Apple Intelligence limits start - The Mac Observer
post: Android 17 is the first since 3.x to add new APIs without releasing to the AOSP

=== 2026-09-19 17:00:00+00:00 | volume spike | 18 posts/h (z=4.0)
news: watchOS 27 eliminates this beloved Apple Watch feature - Yahoo Tech
news: Apple Intelligence bundles Home app alerts by default: here's the off switch - The Mac Observer
news: visionOS 27 brings Siri AI to Vision Pro, even in the EU where iPhone misses out - mixed-news.com
post: Android 17 is the first since 3.x to add new APIs without releasing to the AOSP

=== 2026-09-19 18:00:00+00:00 | engagement spike | 10 posts/h (z=1.3)
news: Top Stories: iOS 27 and macOS Golden Gate Out Now, Siri AI Waitlist, and More - MacRumors
news: New Android Feature Helps You Keep Track Of Important Items, No Tag Needed - SlashGear
news: Apple's Siri AI troubles prove Samsung's messy multi-AI strategy was right - SamMobile
post: Android 17 is the first since 3.x to add new APIs without releasing to the AOSP

=== 2026-09-19 19:00:00+00:00 | engagement spike | 8 posts/h (z=0.7)
news: Top Stories: iOS 27 and macOS Golden Gate Out Now, Siri AI Waitlist, and More - MacRumors
news: New Android Feature Helps You Keep Track Of Important Items, No Tag Needed - SlashGear
news: Apple's Siri AI troubles prove Samsung's messy multi-AI strategy was right - SamMobile
post: Android 17 is the first since 3.x to add new APIs without releasing to the AOSP

=== 2026-09-20 06:00:00+00:00 | volume spike | 16 posts/h (z=3.0)
news: iOS 27 makes your iPhone faster in these 30+ ways - 9to5Mac
news: Apple releases Golden Gate update with new Siri AI for Macs - The Daily Star
news: iCloud Photos Sync Immediately in iOS 27: Here's what it does and when it pauses - The Mac Observer
post: GrapheneOS Isn't Happy With Google Over Pixel's Widening Head Start Google's September 2026 Pixel Update Bulletin contains patches beyond what's in th

```

=== 2026-09-20 07:00:00+00:00 | volume spike | 21 posts/h (z=4.7)
news: Apple releases Golden Gate update with new Siri AI for Macs - The Daily Star
news: iCloud Photos Sync Immediately in iOS 27: Here's what it does and when it pauses - The Mac Observer
news: How to report a bad Siri AI answer to Apple with Report a Concern - The Mac Observer
post: GrapheneOS Isn't Happy With Google Over Pixel's Widening Head Start Google's September 2026 Pixel Up
date Bulletin contains patches beyond what's in th

=== 2026-09-20 09:00:00+00:00 | volume + engagement spike | 23 posts/h (z=4.8)
news: iCloud Photos Sync Immediately in iOS 27: Here's what it does and when it pauses - The Mac Observer
news: How to report a bad Siri AI answer to Apple with Report a Concern - The Mac Observer
news: How to rate photos with stars and add keywords in iOS 27 Photos - The Mac Observer
post: GrapheneOS Isn't Happy With Google Over Pixel's Widening Head Start Google's September 2026 Pixel Up
date Bulletin contains patches beyond what's in th

=== 2026-09-15 00:00:00+00:00 | sentiment shift | net +16% -> -26% (Holm p=0.000)
news: Jamf Announces Same-Day Support For macOS Golden Gate 27, iOS 27 And Other OS 27 Releases - Yahoo Fi
nance
news: Jamf Announces Same-Day Support For macOS Golden Gate 27, iOS 27 And Other OS 27 Releases - Business
Wire
news: Apple releases new parental controls with iOS 27, iPadOS 27, and macOS 27 - BetaNews
post: Tell HN: iOS 27 does not allow Apple Intelligence to be disabled I&#x27;m surprised that this didn&#
x27;t get more attention during the Beta period.<p

```

```

In [21]: # Which words rise fastest around each shift: compares the 12h after against the 12h before.
def surge_terms(ts, hours=12, k=10):
    before = scored[(scored.created_utc >= ts - pd.Timedelta(hours=hours)) & (scored.created_utc < ts)]
    after = scored[(scored.created_utc >= ts) & (scored.created_utc < ts + pd.Timedelta(hours=hours))]
    if len(before) < 20 or len(after) < 20:
        return []
    v = CountVectorizer(ngram_range=(1, 2), min_df=3, stop_words=STOPWORDS)
    B = v.fit_transform(before.text_full.str.lower())
    voc = np.array(v.get_feature_names_out())
    A = CountVectorizer(vocabulary=v.vocabulary_, ngram_range=(1,
2)).fit_transform(after.text_full.str.lower())
    b = np.asarray(B.sum(0)).ravel() / max(len(before), 1)
    a = np.asarray(A.sum(0)).ravel() / max(len(after), 1)
    lift = (a + 1e-4) / (b + 1e-4)
    keep = (np.asarray(A.sum(0)).ravel() >= 5)
    return list(voc[keep][np.argsort(-lift[keep])[:k]])

surges = {}
for _, r in SIG.head(3).iterrows():
    surges[str(r["to"])] = surge_terms(r["to"])
for _, r in spikes.head(3).iterrows():
    surges[str(r.name)] = surge_terms(r.name)
METRICS["surge_terms"] = surges
for k, v in surges.items():
    print(f"{k}: {' '.join(v[:10])}")

```

```

2026-09-15 00:00:00+00:00: beta, photos, feels, experience, new siri, 16, old, release, just, bug
2026-09-09 18:00:00+00:00:
2026-09-09 19:00:00+00:00: siri, intelligence, apple intelligence, ios, 17, ios 27, iphone, apple, ai, 27
2026-09-10 09:00:00+00:00:

```

8. Real-time behaviour: two snapshots

The collector was run twice, and every row records the moment it was fetched. Comparing the snapshots shows the pipeline doing what a monitoring system must do: pick up posts that did not exist at the previous run, and update the picture without re-processing everything.

```

In [22]: if "collected_at" in scored.columns and scored.collected_at.notna().any():
    # The fetch stamp was added for the second run, so rows carrying one are exactly
    # the rows the second snapshot contributed.
    snap = pd.to_datetime(scored.collected_at, format="mixed", utc=True, errors="coerce")
    runs = snap.dropna().dt.floor("h").value_counts().sort_index()
    new_rows = scored[snap.notna()]
    first_run = runs.index.min()

```

```

METRICS["snapshots"] = {
    "runs": {str(k): int(v) for k, v in runs.items()},
    "rows_added_by_second_run": int(len(new_rows)),
    "new_row_sentiment": new_rows.sentiment.value_counts().to_dict() if len(new_rows) else {},
    "newest_post_first_snapshot": str(scored[snap.isna()].created_utc.max()),
    "newest_post_second_snapshot": str(scored.created_utc.max()),
    "reddit_comments_total": int((scored.source.eq("reddit") & scored.kind.eq("comment")).sum()),
}
print(json.dumps(METRICS["snapshots"], indent=2)[:800])
if len(new_rows):
    display(new_rows[["created_utc", "source", "sentiment", "confidence", "text_full"]].head(8))
else:
    METRICS["snapshots"] = {"note": "single snapshot"}
    print("Only one snapshot present in this dataset.")

```

```

{
  "runs": {
    "2026-09-20 10:00:00+00:00": 115,
    "2026-09-20 11:00:00+00:00": 118
  },
  "rows_added_by_second_run": 233,
  "new_row_sentiment": {
    "Negative": 110,
    "Neutral": 91,
    "Positive": 32
  },
  "newest_post_first_snapshot": "2026-09-20 09:33:18+00:00",
  "newest_post_second_snapshot": "2026-09-20 10:44:03.162000+00:00",
  "reddit_comments_total": 41
}

```

	created_utc	source	sentiment	confidence	text_full
7	2026-09-08 12:46:35+00:00	reddit	Negative	0.6230	Can one of the features be separating the phot...
11	2026-09-09 01:17:01+00:00	reddit	Negative	0.5039	Why did you do a weird AI Fact check? The info...
12	2026-09-09 01:20:15+00:00	reddit	Negative	0.5446	I spent time typing this up myself, definitely...
15	2026-09-09 09:30:01+00:00	reddit	Negative	0.6395	I am running the ios 27 beta and don't see thi...
25	2026-09-09 18:25:20+00:00	reddit	Negative	0.8329	A lot of the articles are making a big deal ab...
26	2026-09-09 18:25:31+00:00	reddit	Neutral	0.6363	So will this be available in EU? Since it has ...
27	2026-09-09 18:26:19+00:00	reddit	Neutral	0.5657	The redesigned Health app will be available on...
31	2026-09-09 18:49:25+00:00	reddit	Neutral	0.5015	iOS 27 RC is available now

9. Live refresh check

The pipeline is not a one-off snapshot: the same collector functions can be called again at any moment to pull the newest posts and score them with the same model. The cell below fetches live posts published since the dataset was saved, which is how the Social Engine would run continuously.

```

In [23]: try:
sys.path.insert(0, str(ROOT))
from Entropy_collect_data import mastodon_tag
fresh = pd.DataFrame(sum((mastodon_tag(inst, tag, 1) for inst, tag in
                        [("mastodon.social", "ios27"), ("mastodon.social", "apple"),
                        ("mastodon.social", "android17"), ("mstdn.social", "ios27")])), [])

if len(fresh):
    fresh["created_utc"] = pd.to_datetime(fresh.created_utc, format="mixed", utc=True)
    fresh = fresh[fresh.created_utc > scored.created_utc.max()]
if len(fresh):
    fresh["text_full"] = fresh.text.astype(str)
    pr = engine.predict_proba(fresh.text_full.tolist())
    fresh["sentiment"] = np.array(engine.labels)[pr.argmax(1)]
    METRICS["live_refresh"] = {"new_posts": int(len(fresh)), "checked_at": str(pd.Timestamp.utcnow()),
                              "sentiment": fresh.sentiment.value_counts().to_dict()}
    display(fresh[["created_utc", "source_detail", "sentiment", "text_full"]].head(8))
else:
    METRICS["live_refresh"] = {"new_posts": 0, "checked_at": str(pd.Timestamp.utcnow())}

```

```

        print("No posts newer than the saved dataset were available at refresh time.")
    except Exception as exc:
        METRICS["live_refresh"] = {"error": f"{type(exc).__name__}: {exc}"}
        print("Live refresh unavailable:", exc)

```

	created_utc	source_detail	sentiment	text_full
40	2026-09-20 11:58:50+00:00	mastodon.social	Negative	Don't Be Nice Article URL: https://roe.dev/bl...
41	2026-09-20 11:55:18+00:00	mastodon.social	Neutral	アプリ、待ちましたです トランプ氏の野望は月まで届く？ 地名変更で「遺産」狙い 追従拒否の...
42	2026-09-20 11:55:14+00:00	mastodon.social	Neutral	iPhoneがヒトにとって持つ意味と、垂人にとって持つ意味は同じなのでしょうか 「iPhon...
43	2026-09-20 11:53:51+00:00	mastodon.social	Negative	Welcome to Dogs Anonymous, the Online Support ...
44	2026-09-20 11:48:50+00:00	mastodon.social	Negative	Thermacell LIV 2.0 Review: Taking My Yard Back...
45	2026-09-20 11:43:50+00:00	mastodon.social	Neutral	Show HN: AI Facial Attractiveness Model Align...
46	2026-09-20 11:38:50+00:00	mastodon.social	Positive	I've tested dozens of wireless chargers — this...
47	2026-09-20 11:33:50+00:00	mastodon.social	Negative	Putin casts online vote using unlicensed Windo...

10. Save analysis outputs

```

In [24]: scored.to_csv(SCORED, index=False)
bins.round(4).to_csv(OUT / "Entropy_round3_sentiment_bins.csv")
vol.round(3).to_csv(OUT / "Entropy_round3_activity_hourly.csv")
ents.round(4).to_csv(OUT / "Entropy_round3_entity_sentiment.csv", index=False)
(OUT / "Entropy_round3_metrics.json").write_text(
    json.dumps(METRICS, indent=2, default=lambda o: o.item() if hasattr(o, "item") else str(o)),
    encoding="utf-8")
for p_ in sorted(OUT.glob("*") + sorted(IMG.glob("*"))):
    print(f"{p_.relative_to(ROOT)} {p_.stat().st_size/1024:,.0f} KB")

```

```

outputs\Entropy_round3_activity_hourly.csv 13 KB
outputs\Entropy_round3_annotation_sample.csv 78 KB
outputs\Entropy_round3_entity_sentiment.csv 1 KB
outputs\Entropy_round3_metrics.json 48 KB
outputs\Entropy_round3_scored_posts.csv 1,757 KB
outputs\Entropy_round3_sentiment_bins.csv 1 KB
outputs\Entropy_round3_validation_sample.csv 75 KB
images\Entropy_r3_activity_spikes.png 183 KB
images\Entropy_r3_entity_sentiment.png 118 KB
images\Entropy_r3_entity_sentiment_timeline.png 149 KB
images\Entropy_r3_entity_timeline.png 194 KB
images\Entropy_r3_platform_engagement.png 69 KB
images\Entropy_r3_sentiment_timeline.png 132 KB
images\Entropy_r3_volume_timeline.png 51 KB

```